



Faculty of Engineering and Materials Science
German University in Cairo

Smart Task & Quality Management System

A thesis submitted in partial fulfilment of the requirements for the degree of
Bachelor of Science in Computer Science and Engineering

By

Ahmed Hassan Abdelbarr

Supervised by

Wael Zakaria Abdallah

June 6, 2026

This is to certify that:

- (i) the thesis comprises only my original work toward the Bachelor Degree of Science (B.Sc.) at the German University in Cairo (GUC),
- (ii) due acknowledgment has been made in the text to all other material used

Ahmed Hassan Abdelbarr
June 6, 2026

Acknowledgments

I would like to express my sincere gratitude to my supervisor, **Associate Professor Wael Zakaria Abdallah**, for his guidance, constructive feedback, and continuous support throughout the development of this project and the writing of this thesis.

I am grateful to the Faculty of Engineering and Materials Science at the German University in Cairo for providing a stimulating academic environment and the resources necessary to carry out this work.

My heartfelt thanks go to my family for their unwavering encouragement and patience throughout my studies.

Finally, I extend my appreciation to the open-source communities behind the technologies used in this project—including Microsoft .NET, React, Tauri, Rust, MongoDB, and the Tailwind CSS team—whose publicly available documentation and tooling made this work possible.

List of Publications

No publications arose from this work at the time of submission.

Abstract

Modern knowledge workers increasingly depend on digital task management tools to organise work, coordinate with teams, and meet deadlines. However, most contemporary solutions are cloud-centric, making them unavailable in low-connectivity environments, and none provide a tightly integrated AI assistant that understands the full project context.

This thesis presents *TaskFlow*, an offline-first desktop task and quality management application built with a hybrid architecture. The system pairs a local SQLite database for primary persistence with a MongoDB relay layer for cross-device data synchronisation when connectivity is available. The backend is implemented in ASP.NET Core (.NET 10) using a repository pattern, Entity Framework Core, local-identity middleware, and SignalR for real-time notifications. The frontend is built with React 19 and TypeScript, packaged as a cross-platform desktop application.

TaskFlow provides five distinct task visualisation modes—list, Kanban board, table, Gantt chart, and calendar—alongside project management, team collaboration, real-time direct and group messaging, and a calendar with event scheduling. A key contribution is the embedded AI assistant powered by the Mistral API, offering three specialised modes: a general productivity assistant, a code helper, and a document scanner using optical character recognition. The assistant streams responses incrementally using Server-Sent Events, providing a responsive conversational experience.

An outbox-pattern offline synchronisation engine ensures that operations performed without network access are reliably replayed once connectivity is restored, with progress broadcast to the client via SignalR. The desktop application is packaged with Tauri v2, a Rust-based framework that uses the operating system’s native WebView, resulting in an installer size of approximately 35 MB and cold-start time of approximately 1.5 seconds. The application is distributed as a self-contained NSIS or MSI installer for Windows, with the .NET runtime embedded so no prior installation is required.

Evaluation shows that TaskFlow successfully delivers a full-featured, offline-capable task management environment with integrated AI assistance, addressing a clear gap in the existing landscape of productivity desktop software.

Keywords: task management, offline-first, SQLite, MongoDB, ASP.NET Core, React, Tauri, WebView2, AI assistant, Mistral, real-time synchronisation, SignalR.

Contents

Acknowledgments	3
List of Publications	4
Abstract	5
List of Abbreviations	10
List of Figures	11
List of Tables	13
1 Introduction	14
1.1 Background and Motivation	14
1.2 Problem Statement	15
1.3 Aims and Objectives	16
1.4 Scope and Limitations	17
1.5 Thesis Overview	18
2 Literature Review	19
2.1 Task Management and Productivity Software	19
2.2 Desktop Application Frameworks	20
2.3 Offline-First Architecture	21
2.4 Hybrid Database Architectures	22
2.5 Real-Time Communication Patterns	23
2.6 Authentication and Security	24
2.7 Large Language Models and AI-Assisted Productivity	25
2.8 Project Visualisation Paradigms	26
2.9 Summary	26
3 Methodology	28

3.1	Requirements Analysis	28
3.2	System Architecture Overview	29
3.3	Data Layer Design	31
3.3.1	Dual-Store Strategy	31
3.3.2	Synchronisation Model	31
3.3.3	Entity Model	31
3.3.4	MongoDB Collection Design	34
3.4	Backend Design	36
3.4.1	Authentication Service	36
3.4.2	Task and Project Management	36
3.4.3	Real-Time Notification System	37
3.4.4	Messaging System	37
3.4.5	Offline Synchronisation Engine	37
3.4.6	AI Integration	38
3.5	Frontend Design	39
3.5.1	Application Structure	39
3.5.2	Task Visualisation Views	39
3.5.3	Offline-Aware User Interface	40
3.5.4	AI Chatbot Interface	41
3.6	Desktop Packaging	41
3.6.1	Native Shell	41
3.6.2	Sidecar Backend Management	42
3.6.3	Window Lifecycle	42
3.6.4	Plugin Architecture	42
3.6.5	Security Configuration	42
3.6.6	Build Pipeline	42
3.7	Testing Strategy	43
4	Results	44
4.1	System Features Overview	44
4.2	Authentication and User Management	46
4.3	Task and Project Management	48
4.4	Task Visualisation Modes	51
4.5	Team Collaboration and Messaging	55
4.6	Offline Synchronisation	56
4.7	AI Chatbot	57
4.8	Calendar and Dashboard	59
4.9	Notifications and Settings	60
4.10	Cross-Platform Distribution	61
4.11	Performance Observations	62

5	Conclusion	63
5.1	Summary of Work	63
5.2	Technical Contributions	64
5.3	Lessons Learned	65
5.4	Limitations	66
6	Future Work	68
6.1	Short-Term Improvements	68
6.2	Long-Term Enhancements	69
	Appendix	71
A	API Reference	72
B	Database Schema	74
C	Configuration Reference	77
	References	81

List of Abbreviations

List of Figures

1.1	The TaskFlow desktop application showing the navigation sidebar, dashboard metric cards, mini-calendar, and Task Line widget.	15
3.1	Entity Relationship Diagram for the SQLite Database.	33
3.2	Entity Relationship Diagram for the MongoDB Database.	35
3.3	Kanban view of the Tasks module with five colour-coded status columns and drag-and-drop support.	40
4.1	Dashboard after a task assignment, showing updated metric cards, activity feed, and Task Line widget.	46
4.2	Registration form with FluentValidation-validated identity fields and optional profile fields persisted to <code>AppUser</code>	47
4.3	Login screen that posts credentials to the authentication endpoint and caches the user identity in React state.	48
4.4	New-task editor in its initial state showing all required and optional fields with the <i>Smart Auto Fill</i> panel at the top.	49
4.5	Task editor after <i>Smart Auto Fill</i> populated fields from a natural-language prompt via the Mistral OCR model.	50
4.6	Default view with tasks partitioned into temporal groups (<i>Overdue, Today, This Week, Later, Completed</i>).	51
4.7	Kanban view with a task in the <i>To Do</i> column, colour-coded status headers, search and filter toolbar.	52
4.8	Table view presenting tasks as rows with sortable columns and colour-coded priority and status pill badges.	53
4.9	Gantt view showing task bars on a rolling 10-week forward timeline with a current-date marker.	54
4.10	Calendar view with a mini-calendar sidebar and a monthly grid of task and event blocks in adjacent lanes.	55
4.11	AI Assistant landing surface with three mode toggles (<i>General, Coder, Scanner</i>) and a starter-prompt grid.	57

4.12	AI Assistant in <i>Coder</i> mode producing a syntax-highlighted code block via the Mistral API streamed response.	58
4.13	Dedicated <i>Calendar</i> page with a full-month time-grid view and the <i>New Event</i> creation form.	59
4.14	Notifications page with unread items delivered via SignalR, <i>Mark all as Read</i> and <i>Delete All</i> actions.	60
4.15	Settings page (Profile tab) with navigation rail, avatar upload, and editable personal-information fields.	61

List of Tables

2.1	Feature comparison of popular task-management tools vs. TaskFlow.	20
2.2	Comparison of Electron and Tauri v2 desktop application frameworks.	21
4.1	Functional requirement fulfilment matrix.	45
4.2	Informal response-time observations for key operations.	62
A.1	REST API controllers and primary endpoints.	73
B.1	SQLite entity summary.	75
B.2	MongoDB collection descriptions.	76
C.1	appsettings.json section reference.	78
C.2	Supported environment variables.	79
C.3	Key tauri.conf.json configuration (Tauri build).	80

Chapter 1

Introduction

1.1 Background and Motivation

The way people organise their work has changed fundamentally over the last decade. The proliferation of distributed teams, freelance arrangements, and hybrid work models has driven enormous growth in the market for digital task-management and project-collaboration software [1]. Tools such as Trello, Asana, Jira, and Notion have become everyday staples in software development teams, enabling members to track progress, share context, and coordinate across time zones without the overhead of physical meetings.

Despite this progress, the dominant generation of collaboration tools shares a critical architectural assumption: *a reliable, always-available internet connection*. Every create, read, update, and delete operation is routed through a remote cloud server. When connectivity degrades—as it does on mobile hotspots, in aeroplanes, in rural areas, or during network maintenance—these applications either display error banners or silently discard user input. For a professional whose productivity depends on continuous access to their task list, such fragility is unacceptable.

A second limitation is the growing gap between the capabilities of modern Large Language Models (LLMs) and the AI integrations available in mainstream productivity tools. Although several task managers have begun experimenting with AI-generated summaries, none provides a fully embedded, context-aware AI assistant that understands the user’s current project state and can assist with both natural-language planning and technical code generation within the same interface [2].

These two observations—the unreliability of purely cloud-centric architectures and the absence of deep AI integration—form the motivation for *TaskFlow: a Smart Task & Quality Management System* designed from the ground up as an offline-first desktop application with an embedded multi-mode AI assistant.

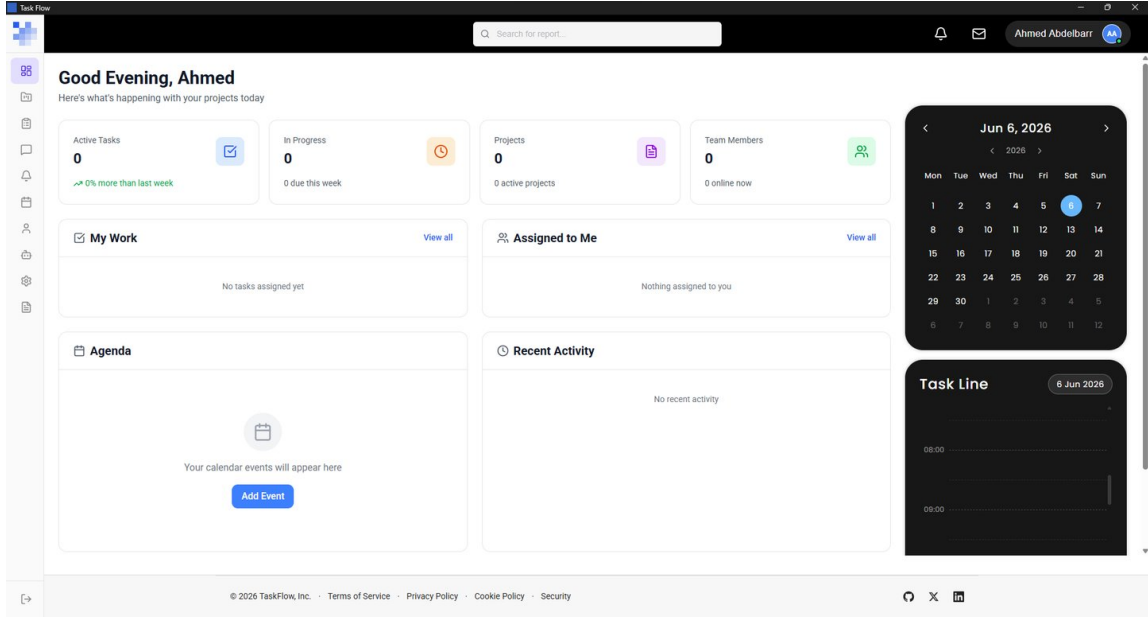


Figure 1.1: The TaskFlow desktop application showing the navigation sidebar, dashboard metric cards, mini-calendar, and Task Line widget.

1.2 Problem Statement

The central problem addressed by this project can be stated as follows:

Existing task-management tools treat internet connectivity as a prerequisite rather than an enhancement. They provide no graceful degradation path, no local data store, and no mechanism to synchronise divergent offline edits once connectivity is restored. Furthermore, their AI capabilities, where present, are superficial and disconnected from the user's live project data.

More concretely, a review of widely-used tools reveals the following gaps.

1. **No offline persistence.** Cloud-first tools store all state remotely. A user working without connectivity cannot create tasks, update statuses, or leave comments; changes attempted offline are either silently lost or surfaced as errors.
2. **No transparent synchronisation.** Even tools that offer limited offline caching typically give users no visibility into which changes are pending, in progress, or confirmed. The sync lifecycle is a black box, eroding trust in the accuracy of the displayed data.
3. **Fragmented toolchains.** Developers and project managers routinely switch between a task manager, a code assistant, a calendar, a messaging client, and a notification centre. There is no single, cohesive desktop application that integrates all of these capabilities into one consistent experience.

4. **Shallow AI integration.** AI features in existing tools are limited to simple text generation (e.g., auto-writing a task description). None provides a conversational assistant capable of code generation, document analysis, or multimodal input—all within the context of the user’s own project data.

1.3 Aims and Objectives

The primary aim of this project is to design, implement, and evaluate a cross-platform desktop task-management application that operates reliably regardless of network availability, while embedding a powerful AI assistant and a full suite of collaboration features.

The following specific objectives guide the work:

1. **Offline-first operation.** Every core feature—task creation and editing, project management, calendar events, notifications, and messaging—must function fully without an active internet connection. Data must be persisted locally in a SQLite database for performance, with MongoDB serving as the single source of truth for synchronisable data across devices.
2. **Reliable, transparent synchronisation.** When connectivity is available, locally generated changes must be propagated to a MongoDB relay database using a transactional outbox pattern that guarantees at-least-once delivery. The user must be able to see the current connectivity status, the count of pending operations, and the real-time progress of any active sync pass through a dedicated UI indicator.
3. **Integrated multi-mode AI assistant.** A conversational AI assistant backed by the Mistral API must be embedded directly in the application. It must support three operational modes: a general-purpose chat mode, a code-generation mode powered by Codestral, and a document-analysis mode capable of processing uploaded images and PDF files via optical character recognition (OCR). Responses must be streamed to the UI via Server-Sent Events (SSE) to provide a responsive, real-time interaction feel.
4. **Five complementary task visualisation views.** The task module must offer five distinct views catering to different cognitive styles and workflows: a smart-grouped default list, a Kanban board with drag-and-drop, a tabular data-grid, a Gantt-style timeline, and a calendar time-grid.
5. **Cross-platform desktop distribution.** The completed application must be packaged as a native installer for Windows (NSIS and MSI) using Tauri v2, bundling both the .NET backend and the React frontend into a single self-contained executable that requires no external runtime installation by the end user. The Tauri framework was chosen for its significantly smaller installer size and faster cold-start performance, achieved by leveraging the operating system’s native WebView rather than bundling a browser engine.

6. **Comprehensive collaboration feature set.** The system must include team management, one-to-one and group messaging with file attachments, real-time notifications via SignalR, calendar events, a dashboard with productivity statistics, and user-profile settings including security and preference controls.

1.4 Scope and Limitations

In Scope

The following capabilities were designed and implemented as part of this project:

- Full user authentication with local-identity middleware, bcrypt password hashing, and local file-based password-reset workflow.
- Task management with priority levels (Low, Medium, High), status workflow (To Do, In Progress, Review, Completed, Overdue), due-date tracking, starring, and assignment.
- Five task visualisation views: Default, Kanban, Table, Gantt, and Calendar.
- Project and team management with role-based membership and team invitations synchronised through MongoDB.
- One-to-one direct messaging and group chats, both supporting file attachments and soft-delete.
- In-application and email reminders triggered by configurable fire-at times.
- Automated due-date warnings fired 24 hours and 1 hour before deadline.
- Real-time push notifications delivered over SignalR.
- An AI chatbot with three modes, SSE streaming, and conversation history.
- Calendar events with a weekly time-grid view.
- A dashboard with live statistics, activity feeds, and a calendar widget.
- Offline-first SQLite persistence and an outbox-based MongoDB synchronisation subsystem.
- Cross-platform desktop packaging using Tauri v2, producing a self-contained Windows installer with a significantly reduced footprint.

Limitations

The following items were explicitly excluded from the current implementation:

- **Mobile clients.** iOS and Android applications were not developed; the system targets desktop platforms only.
- **Multi-user conflict resolution.** The synchronisation subsystem uses a last-write-wins strategy. Collaborative real-time editing with Conflict-free Replicated Data Types (CRDTs) [3] is a known future enhancement.
- **Hierarchical sub-tasks.** A sub-task UI scaffold is present in the Table view but the backend persistence layer for sub-tasks was not implemented in this iteration.
- **SaaS deployment and billing.** The application is designed for single-user or small-team use on a local machine; multi-tenant cloud hosting and subscription management are out of scope.
- **Externally managed AI credentials.** The Mistral API key is currently supplied at build time. Secure external key management (e.g., system keychain or environment-variable injection) is identified as a known improvement area in Chapter 5.

1.5 Thesis Overview

The remainder of this thesis is structured as follows.

Chapter 2 — Literature Review and Background surveys the academic and technical context of the work, covering task-management paradigms, offline-first architecture, hybrid database design, real-time communication protocols, authentication, and the state of LLM-based AI assistants.

Chapter 3 — Methodology and System Design presents the full system architecture, the rationale behind key design decisions, the data model, and a detailed walkthrough of the back-end, frontend, and desktop-packaging implementations.

Chapter 4 — Results and System Evaluation demonstrates the implemented features through functional descriptions and user-interface walkthroughs, assessing the system against each objective stated in Section 1.3.

Chapter 5 — Conclusion reflects on the outcomes of the project, summarises its technical contributions, and acknowledges known limitations.

Chapter 6 — Future Work outlines a prioritised roadmap of short-term improvements and long-term enhancements that could extend the system.

Appendices provide supplementary technical reference material including the REST API endpoint catalogue, the complete database schema, and the application configuration reference.

Chapter 2

Literature Review

This chapter surveys the academic literature and technical background that supports the design decisions made in TaskFlow. The review spans seven areas: existing task-management tools and their limitations, desktop application frameworks, offline-first architecture patterns, hybrid database designs, real-time communication protocols, authentication and security, and the evolution of Large Language Models (LLMs) as productivity aids. It concludes with a brief treatment of project-visualisation paradigms.

2.1 Task Management and Productivity Software

Task management software encompasses a broad spectrum of tools—from simple personal to-do lists to enterprise project-management suites. The most widely cited personal productivity framework is David Allen’s *Getting Things Done* (GTD) [1], which argues that externalising tasks into a trusted system frees cognitive resources for execution rather than recall. Modern software tools attempt to operationalise this principle in varying ways.

Trello implements the Kanban paradigm [4] using cards on boards with user-defined columns. Its simplicity has made it popular for personal and small-team use, but it provides no offline capability; every drag-and-drop operation requires an active connection. **Asana** and **Jira** extend the model with structured project hierarchies, sprint planning, and reporting dashboards, but both are unambiguously cloud-first and offer only read-only offline caching at best. **Notion** introduces a flexible block-based editor that unifies notes, databases, and tasks in one surface, yet it shares the same connectivity dependency.

Table 2.1 summarises the key feature dimensions of these tools relative to TaskFlow.

Table 2.1: Feature comparison of popular task-management tools vs. TaskFlow.

Feature	Trello	Asana	Jira	Notion	TaskFlow
Offline operation	×	×	×	×	✓
Local persistence	×	×	×	Limited	✓
Transparent sync	×	×	×	×	✓
Embedded AI assistant	Limited	Limited	Limited	Limited	✓
Desktop installer	×	×	×	✓	✓
Kanban view	✓	✓	✓	✓	✓
Gantt / timeline view	×	✓	✓	×	✓
Calendar view	×	✓	✓	✓	✓
Integrated messaging	×	×	×	×	✓

The pattern that emerges from this survey is consistent: each tool excels in one or two dimensions but requires multiple third-party integrations to achieve a comparable feature breadth to TaskFlow, and none treats offline reliability as a design priority.

2.2 Desktop Application Frameworks

The choice of desktop application framework has long-term implications for developer productivity, binary size, platform reach, and the ability to reuse existing web technology investments.

Native frameworks such as WinForms, WPF (Windows), Cocoa (macOS), and GTK (Linux) offer tight OS integration and the best runtime performance but impose a per-platform development effort and cannot reuse the React/TypeScript frontend code already written for TaskFlow.

Tauri [5] is a Rust-based framework for building desktop applications using the operating system’s native WebView rather than bundling a browser engine. On Windows, this WebView is provided by Microsoft WebView2, which is distributed as part of Windows 10 (version 1803 and later) and all versions of Windows 11 [6]. A minimal Tauri application compiles to approximately 600 KB [5], in contrast to the roughly 150 MB baseline of an Electron application. Tauri v2 employs a capabilities-based security model that granularly controls which system APIs each window may invoke [7], and it supports a plugin architecture that provides shell access, file system operations, dialog boxes, and HTTP client functionality. The Rust language used for the Tauri core provides memory safety guarantees and thread safety at compile time [8].

For TaskFlow, the ASP.NET Core backend is bundled as a *sidecar* binary [9]: a self-contained executable that the Tauri application spawns as a child process via the shell

plugin. The sidecar communicates its dynamically assigned port number by writing `TASKFLOW_BACKEND_READY:{address}` to standard output, which the Rust backend manager (`backend.rs`) parses line by line. Once the address is received, the splashscreen window is closed and the main application window is displayed. This stdout-based port negotiation protocol decouples the frontend load from the backend start time, allowing the UI to display a splash state until the server is confirmed ready.

Electron [10], developed by GitHub and now maintained by the OpenJS Foundation, embeds a Chromium browser and a Node.js runtime in a single cross-platform binary. The web content layer communicates with the Node.js main process via inter-process communication (IPC). This architecture allows a complete React + TypeScript frontend to be reused without modification. Table 2.2 summarises the key differences between Electron and Tauri. The TaskFlow desktop shell uses Tauri v2, chosen for its significantly smaller binary footprint, OS-native WebView rendering, and declarative capabilities-based security model [7].

Table 2.2: Comparison of Electron and Tauri v2 desktop application frameworks.

Characteristic	Electron	Tauri v2
Rendering engine	Bundled Chromium	OS native WebView
System language	Node.js (JavaScript)	Rust
Minimum bundle size	~150 MB	~600 KB (app only)
IPC mechanism	contextBridge / preload	Tauri commands (<code>invoke</code>)
Security model	Manual (sandbox flags)	Declarative capabilities system
WebView on Windows	Chromium (bundled)	WebView2 (OS-provided)
Memory footprint	High (Chromium process)	Low (native WebView)
Frontend technology	Any web technology	Any web technology
Plugin ecosystem	npm packages	Tauri plugins (Rust + JS)

2.3 Offline-First Architecture

The term *offline-first* was popularised in the context of progressive web applications and mobile clients, but the underlying principle applies equally to desktop software: *design the system to function fully without network access, and treat synchronisation as an enhancement rather than a prerequisite*.

Kleppmann’s *Designing Data-Intensive Applications* [11] provides a thorough treatment of the trade-offs involved. In the context of the CAP theorem [12], an offline-first system prioritises *Availability* and *Partition Tolerance* over strict *Consistency*: local writes are always accepted, and consistency is restored asynchronously when connectivity returns.

The Transactional Outbox Pattern

A key challenge in offline-first systems is ensuring that locally committed changes are eventually propagated to the remote store without duplication or loss. The *transactional outbox pattern* [13] addresses this by writing both the business entity change and a corresponding outbox entry in the same local database transaction. A background worker polls the outbox and replays pending entries to the remote system, advancing each entry through a state machine: Pending → Processing → Synced (or Failed after exhausting retries). This design guarantees at-least-once delivery while keeping the application logic decoupled from network availability.

Conflict Resolution

When multiple clients modify the same entity while offline, the system must resolve divergent versions upon reconnection. Conflict-free Replicated Data Types (CRDTs) [3, 14] provide a mathematically principled approach: data structures are designed so that any two states can be merged without coordination. Implementing CRDTs in a production system adds significant complexity; the current TaskFlow implementation adopts a simpler *last-write-wins* strategy (using the UpdatedAt timestamp as the arbitration key), with CRDTs noted as a long-term enhancement.

2.4 Hybrid Database Architectures

SQLite as a Local Store

SQLite [15] is an embedded, serverless, ACID-compliant SQL database engine that stores data in a single file on disk. These properties make it the natural choice for a local persistence layer: the database travels with the application, survives process restarts, and performs all reads and writes without network round-trips.

In TaskFlow, MongoDB serves as the *single source of truth* for synchronisable data. All application features read from and write to SQLite for local performance; changes are propagated to MongoDB asynchronously through the outbox-based sync engine, which does not block the user-visible operation path.

MongoDB as a Cloud Relay

MongoDB [16] is a document-oriented database used as the cloud relay layer for cross-device synchronisation.

In TaskFlow, MongoDB serves as a *synchronisation relay* for the subset of data that must be visible across devices or shared among team members: mirrored tasks, notifications, team

invitations, and user-presence indicators. The document `_id` for each mirrored document is the entity's `SyncId` (a UUID generated at creation time), ensuring idempotent upserts and stable cross-device identity. This dual-database pattern—ACID-safe local SQLite as primary, schemaless MongoDB as asynchronous relay—is a pragmatic alternative to full CRDT-based synchronisation when conflict probability is low and eventual consistency is acceptable.

Entity Framework Core

Entity Framework Core [17] is the Object-Relational Mapper used by TaskFlow's backend. Its provider model allows the same application code to target SQLite locally or SQL Server in production through a single configuration switch.

2.5 Real-Time Communication Patterns

Modern collaborative applications require the server to *push* state changes to connected clients rather than waiting for clients to poll. Three primary mechanisms are relevant to TaskFlow.

WebSockets

The WebSocket protocol [18] establishes a persistent, full-duplex TCP connection between a client and server after an HTTP upgrade handshake. Both parties may send frames at any time without the overhead of repeated HTTP headers. WebSockets are well-suited for low-latency bidirectional communication such as chat messages and live presence indicators.

SignalR

ASP.NET Core SignalR [19] is a higher-level abstraction over WebSockets, Server-Sent Events, and long polling. It provides a hub-based RPC model in which the server can invoke named methods on connected clients, and clients can invoke methods on the server. SignalR automatically negotiates the best available transport and handles reconnection. TaskFlow uses SignalR's `NotificationHub` to broadcast real-time events: connectivity-state changes, synchronisation progress, task reminders, and new notification payloads are all delivered to the React client via a single hub connection.

Server-Sent Events

Server-Sent Events (SSE) is a unidirectional streaming protocol built on standard HTTP: the server holds the response connection open and writes `text/event-stream` chunks as data becomes available [19]. SSE is appropriate for AI response streaming because the communi-

cation is inherently one-directional and does not require the connection upgrade ceremony of WebSockets.

2.6 Authentication and Security

Local Identity Middleware

Because the ASP.NET Core backend runs exclusively on localhost and is accessible only from the local desktop application, TaskFlow does not require a remote authentication server or token-based session management. Instead, a custom local-identity middleware resolves the active user from the SQLite database on every request. The frontend identifies the current user by sending an `X-User-Email` HTTP header; the middleware looks up the corresponding `AppUser` record and constructs a `ClaimsPrincipal` with the user's identity, email, and display name. If no header is present, the middleware falls back to the first user in the database, which is sufficient for a single-user desktop deployment. This approach keeps the authentication layer stateless within a session while avoiding the complexity of JWT issuance, refresh, and revocation in a purely local context.

Password Hashing with bcrypt

Storing passwords as plaintext or with reversible encryption is a critical security failure. The bcrypt algorithm [20] applies an adaptive key-derivation function with a configurable cost factor, deliberately making hash computation slow. As hardware performance improves, the cost factor can be increased to maintain the same brute-force resistance. TaskFlow uses `BCrypt.Net-Next` with the default cost factor to hash passwords at registration and verify them at login, ensuring that even a full database compromise does not directly expose user credentials.

OWASP Top 10 Considerations

The Open Web Application Security Project (OWASP) publishes an annually updated list of the ten most critical web application security risks [21]. The most relevant to TaskFlow are: *Broken Access Control*—mitigated by validating the `X-User-Email` header identity against entity ownership on every protected endpoint; *Cryptographic Failures*—addressed by bcrypt password hashing; *Injection*—mitigated by EF Core's parameterised queries which eliminate raw SQL string concatenation; and *Security Misconfiguration*—acknowledged in the context of the Mistral API key, which should be supplied via an environment variable rather than embedded in source code.

Role-Based Access Control

Ferraiolo and Kuhn’s foundational work on role-based access control (RBAC) [22] defines a model in which permissions are assigned to roles and roles are assigned to users, rather than assigning permissions directly to users. TaskFlow implements a lightweight RBAC model for team membership: each `TeamMember` record carries a `Role` field, and the `Teams` controller enforces that only the team owner may perform administrative operations such as removing members or disbanding the team.

2.7 Large Language Models and AI-Assisted Productivity

The Transformer Architecture

The transformer architecture, introduced by Vaswani et al. [23] in the seminal paper *Attention Is All You Need*, replaced recurrent sequence-to-sequence models with a purely attention-based mechanism. The self-attention operation allows each token in a sequence to attend to every other token in parallel, enabling efficient training on large corpora and strong performance on a wide range of language tasks. This architecture became the foundation for all major LLMs in subsequent years.

GPT-3 and the Emergence of Instruction Following

Brown et al.’s GPT-3 [24] demonstrated that scaling transformer models to 175 billion parameters produced qualitatively new capabilities, including few-shot task completion from natural-language prompts. The subsequent development of instruction-tuned variants established the prompt-response interaction model that underpins modern conversational AI assistants.

Mistral Models

Jiang et al. [2] introduced Mistral 7B, a 7-billion-parameter open-weight model that outperformed LLaMA 2 13B on several benchmarks despite being half the size, achieved through grouped-query attention and sliding-window attention mechanisms. TaskFlow integrates the Mistral API at three model tiers: `mistral-small-latest` for general conversational use; `codestral-latest` for the code-generation mode, a model fine-tuned on a large corpus of programming languages; and `mistral-ocr-latest` for the document-scanner mode, which accepts base64-encoded images and PDF files and returns extracted and structured text. The 20-message conversation history window provides conversational context while bounding the per-request token count.

2.8 Project Visualisation Paradigms

Different users hold different mental models of their work, and no single visualisation paradigm suits all cognitive styles or project types. A mature task-management tool should offer multiple views over the same underlying data.

Kanban boards originated in Toyota’s production system as physical cards (*kanban* = “signal card” in Japanese) that controlled the flow of work between manufacturing stages [4]. Their digital equivalent presents tasks as cards in columns representing workflow stages, giving teams an immediate visual signal of bottlenecks (columns with too many cards) and idle capacity. Drag-and-drop interaction directly maps the physical act of moving a card from one column to the next.

Gantt charts were developed by Henry Gantt in the early twentieth century as project-schedule representations that map tasks onto a horizontal time axis [25]. They are particularly effective for deadline-sensitive planning because the chart immediately communicates which tasks are in progress, which are complete, and which are overdue relative to a calendar.

Calendar views represent tasks as events on a time grid, a mental model familiar to users from personal calendars. A weekly or monthly grid makes it easy to reason about task density and identify overloaded days.

Tabular views surface all task attributes as a structured grid, enabling rapid scanning, sorting, and filtering without the spatial metaphor of Kanban columns or timeline bars. This view is preferred by users who think in terms of data rather than process flow.

TaskFlow provides all four of these paradigms plus a smart-grouped default list (grouping tasks by temporal proximity: Overdue, Today, This Week, Later, Completed), giving each user the freedom to work in the style that best fits their workflow at any given moment.

2.9 Summary

The literature review reveals a clear pattern of unmet needs in the current landscape of task-management software. Existing tools are built on cloud-first assumptions that make them unreliable in low-connectivity environments. The offline-first literature, particularly the transactional outbox pattern described by Fowler [13] and the data-intensive application principles formalised by Kleppmann [11], provides the theoretical basis for a reliable local-first persistence and synchronisation subsystem. The dual-database architecture combining SQLite and MongoDB is a pragmatic application of these principles that avoids the complexity of full CRDT-based conflict resolution while still enabling cross-device data relay.

In the domain of AI integration, the Mistral family of models [2] offers an open-weight, commercially accessible set of capabilities—general language, code generation, and OCR—

that align precisely with the three modes required by the system. Server-Sent Events provide the appropriate transport mechanism for streaming AI responses to the client with minimal protocol overhead.

The desktop application was built with Tauri v2, chosen for its significantly smaller binary footprint, OS-native WebView rendering, and declarative security model. The bcrypt password hashing algorithm and the local-identity middleware together provide a well-understood and academically grounded authentication and security foundation.

The following chapter translates these theoretical foundations into a concrete system design, detailing the architecture, data model, and implementation of each subsystem in TaskFlow.

Chapter 3

Methodology

This chapter describes the methodology used to design and implement TaskFlow. It begins with a structured requirements analysis and a high-level system architecture overview, then examines each major subsystem through the lens of design decisions, architectural trade-offs, and alternative approaches considered. The chapter closes with a discussion of the testing strategy applied throughout development.

3.1 Requirements Analysis

Functional Requirements

The following functional requirements were identified from the project objectives stated in Section 1.3:

1. **FR-01 User Authentication.** The system shall support user registration, login, and password reset. Credentials shall be stored using a salted, one-way hash and protected against replay attacks.
2. **FR-02 Task Management.** Users shall be able to create, read, update, and delete tasks. Each task shall carry a title, optional description, configurable priority, status, optional due date, star flag, and project assignment.
3. **FR-03 Multiple Task Views.** Tasks shall be displayable in five views: a smart-grouped default list, a Kanban board, a tabular grid, a Gantt timeline, and a calendar time-grid.
4. **FR-04 Project and Team Management.** Users shall be able to create projects, invite team members, and assign tasks to projects. Invitations shall be delivered across devices.
5. **FR-05 Messaging.** Users shall be able to exchange one-to-one direct messages and participate in group chats, both with optional file attachments.

6. **FR-06 Notifications and Reminders.** The system shall deliver real-time in-application notifications and optional email reminders at user-configured times. Automated warnings shall fire before task due dates.
7. **FR-07 Calendar Events.** Users shall be able to create, edit, and delete calendar events with temporal boundaries, an optional colour, and an optional meeting link.
8. **FR-08 AI Chatbot.** A conversational AI assistant shall be available in three modes: general conversation, code generation, and document scanning. Responses shall be delivered token-by-token as they are generated. Conversation history shall be persisted between sessions.
9. **FR-09 Offline Operation.** All core task-management features shall function without network access. Locally generated changes shall be queued and synchronised when connectivity is restored.
10. **FR-10 Dashboard.** A dashboard shall display summary statistics, a task-trend indicator, tasks due within the current week, a recent-activity feed, and an embedded calendar widget.
11. **FR-11 Settings.** Users shall be able to update their profile, change their password, and manage notification preferences.

Non-Functional Requirements

1. **NFR-01 Offline Availability.** The application shall start and remain fully operable without network access.
2. **NFR-02 Security.** All API endpoints shall require authentication. Passwords shall not be stored in recoverable form. The application shall be resistant to common web vulnerabilities including SQL injection.
3. **NFR-03 Cross-Platform Distribution.** The packaged installer shall run on Windows, macOS, and Linux without requiring the user to install any additional runtime.
4. **NFR-04 Responsiveness.** UI interactions shall produce visible feedback within 200 ms under normal conditions.
5. **NFR-05 Extensibility.** The backend shall be structured so that new entity types can be added without modifying existing controller logic.

3.2 System Architecture Overview

TaskFlow is structured as a three-tier desktop application embedded in a native operating-system shell.

Tier 1 — Presentation Layer. The user interface is a single-page application built with React and TypeScript. It communicates with the backend exclusively through HTTP REST calls and a persistent WebSocket connection for real-time events. The choice of an SPA architecture over server-side rendering was motivated by the need for a responsive, application-like feel in a desktop context, where page reloads would disrupt the user experience.

Tier 2 — Application Logic Layer. The backend exposes a REST API and a real-time event hub. It follows a layered architecture: HTTP controllers delegate to service classes, which in turn use data-access objects for persistence. A mapping layer projects internal entity models onto response objects at the service boundary, preventing internal representation details from leaking into the API surface. A global error-handling mechanism catches unhandled exceptions and returns a uniform error response, ensuring that no internal state or stack trace is ever exposed to the client. This layered structure was chosen to maintain separation of concerns and to allow individual layers to be tested or replaced independently.

Tier 3 — Data Layer. Two database systems are employed for different purposes. A local embedded database serves as the primary, always-available store that operates without any network dependency. A cloud document database acts as an optional relay used exclusively for cross-device synchronisation and team collaboration features. The application starts and runs normally even when the cloud database is unreachable. This dual-store design was adopted to satisfy the offline-availability requirement (NFR-01) while still enabling multi-device and team workflows.

Desktop Shell. A lightweight native shell launches the compiled backend as a managed child process. It monitors the child's output for a readiness signal before revealing the application window, ensuring that the user interface is never shown before the API is ready to serve requests. Using a native shell rather than bundling a full browser engine reduces the application's disk and memory footprint, as it leverages the operating system's built-in webview component.

Request Lifecycle. A typical authenticated request flows through the layers in sequence: the React component issues an HTTP request with the user's identity — the shell process adds the authenticated user's email to every outgoing request. The backend middleware resolves this identity into a principal object that is carried through the remainder of the request pipeline. The controller delegates to a service, which performs the business logic and persists changes through the data-access layer. If the operation mutates a synchronisable entity, the service additionally triggers the synchronisation engine, which either applies the change directly to the cloud database (if online) or enqueues it for later replay (if offline). Real-time events are broadcast to the connected client via the persistent WebSocket connection independently of the HTTP response path.

3.3 Data Layer Design

3.3.1 Dual-Store Strategy

The central design decision in the data layer was the adoption of two distinct database systems serving complementary roles. A local embedded database (SQLite) provides the primary persistence layer with zero network dependency, satisfying the offline-availability requirement. A cloud document database (MongoDB) acts as a synchronisation relay for cross-device features: team invitations, membership rosters, presence information, and mirrors of synchronisable entities such as tasks and notifications.

This dual-store approach introduces inherent complexity — every write to a synchronisable entity must eventually be reflected in both stores, and the system must handle periods where only one store is reachable. An alternative considered was a cloud-only architecture with a local cache, but this was rejected because it would make the application unusable during network outages, violating NFR-01. Conversely, a local-only architecture would prevent multi-device and team features entirely. The chosen design occupies a middle ground: the cloud store (MongoDB) is authoritative, and the local store is a local cache whose consistency with the source of truth is maintained by an outbox-based synchronisation engine.

3.3.2 Synchronisation Model

Entities that must be replicated to the cloud store are identified at design time. Each such entity carries a globally unique identifier assigned at creation time, which serves as the document key in the cloud database. This design enables idempotent operations: replaying the same synchronisation entry twice produces the same document rather than a duplicate. A timestamp field records the most recent modification and serves as the arbitration key in a last-write-wins conflict-resolution strategy, which was chosen for its simplicity and because concurrent edits to the same entity across different devices are expected to be rare in a personal-task-management context.

A dirty flag on each synchronisable entity indicates whether the local record has been propagated to the cloud. The persistence layer automatically stamps the modification timestamp and resets the dirty flag on any tracked change, eliminating the need for manual bookkeeping in service code.

3.3.3 Entity Model

The application manages eighteen entity types in the local database, grouped into identity, work items, team and social structures, messaging, time-related records, alerts, AI conversation history, and synchronisation metadata. Unique indexes on the global identifiers of synchronisable

types prevent duplicate inserts if the application is restarted during a synchronisation pass. Entity types that are read-only mirrors of cloud documents — such as team memberships and invitations pulled on login — use the cloud document identifier as their local primary key.

Figure 3.1 illustrates the entity relationship diagram for the local SQLite database.

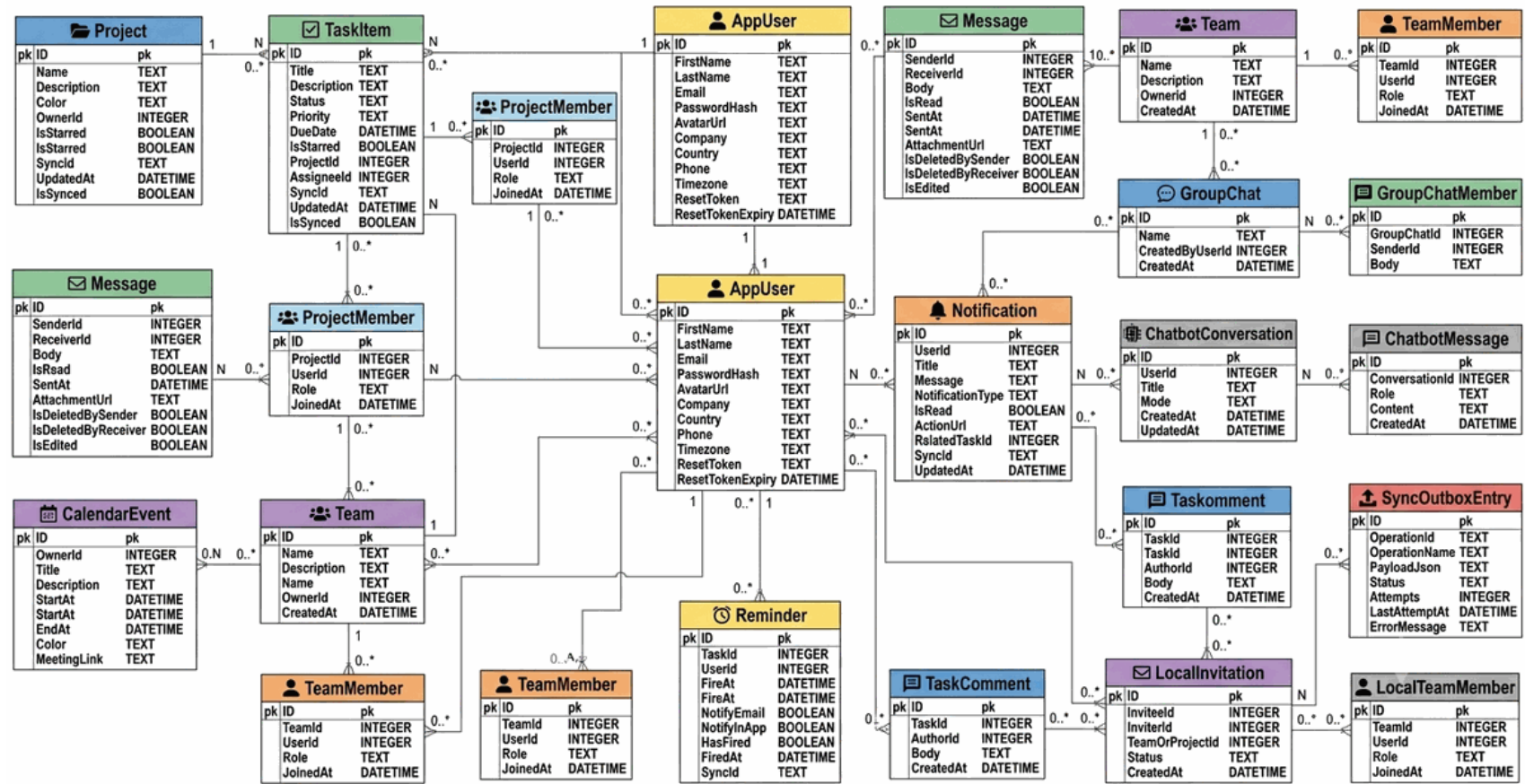


Figure 3.1: Entity Relationship Diagram for the SQLite Database.

3.3.4 MongoDB Collection Design

Five collections are maintained in the cloud database:

- Synchronisation mirrors of the primary work entities (tasks, notifications), keyed by their global identifiers.
- A team-invitations collection indexed by recipient email and status, enabling pending invitations to be delivered to any device the recipient logs into.
- A team-membership collection indexed by team and owner, serving as the roster for team-wide features.
- A user-presence collection for online-status indication across devices.

This minimal cloud schema reflects the design principle that the cloud store exists solely to enable cross-device and team collaboration — it is not a backup or primary store.

Figure 3.2 visualises the collections and their relationships within the MongoDB database.

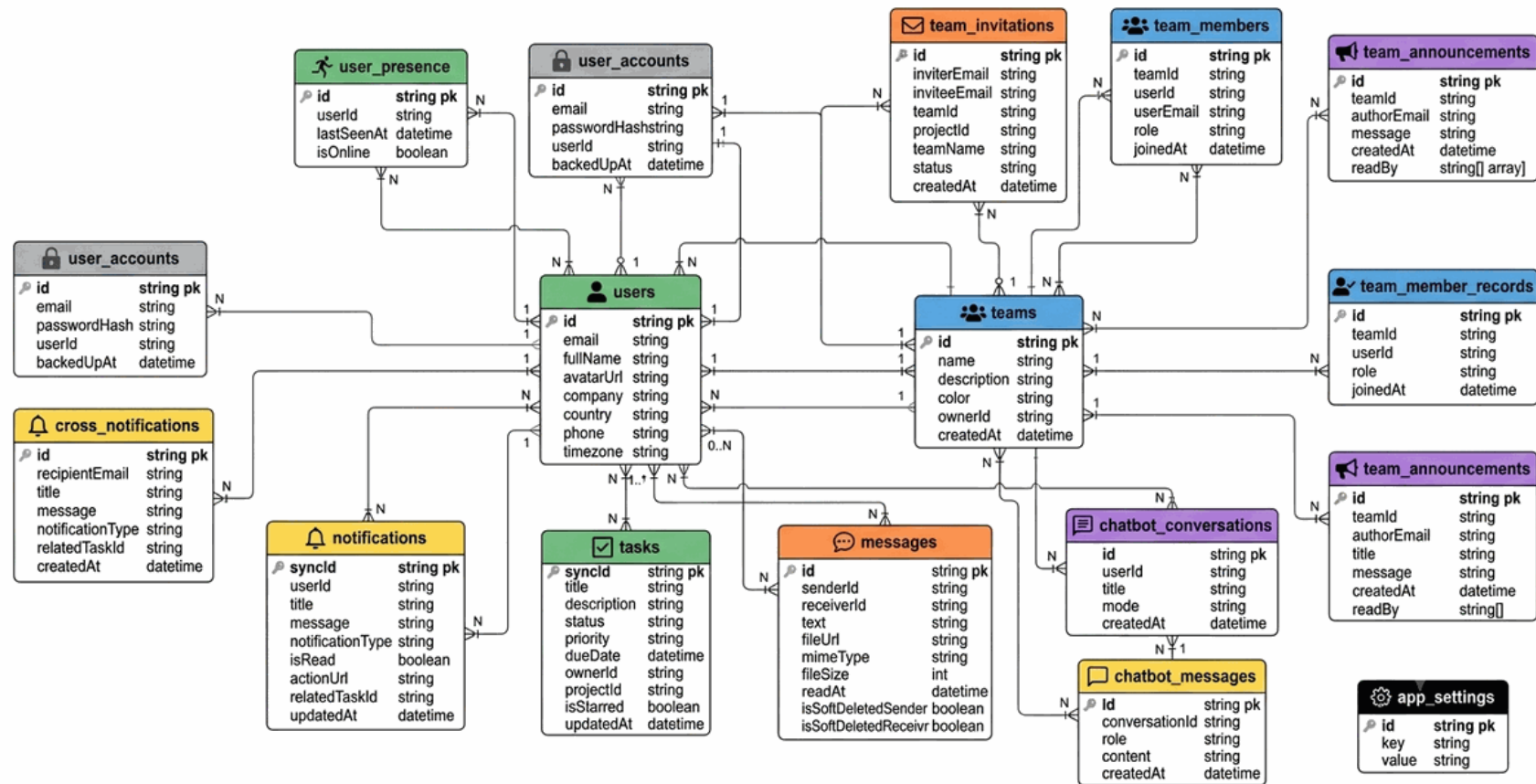


Figure 3.2: Entity Relationship Diagram for the MongoDB Database.

3.4 Backend Design

3.4.1 Authentication Service

User registration follows a three-step process: input validation, credential hashing, and persistence. The choice of a salted, adaptive hash function (bcrypt) for password storage was made to resist brute-force attacks even if the local database file is compromised — a realistic threat model for a desktop application whose data resides on the local filesystem.

Login validates the submitted password against the stored hash and, on success, returns the user's identity profile. No token is issued; the session is managed by middleware that reads the authenticated user's email from each request header and resolves the corresponding identity from the local database. This approach was chosen over a token-based scheme — such as JWT — to eliminate token issuance, refresh, and revocation complexity in a single-user desktop context where the HTTP client and server are always on the same machine and the identity header is set by the trusted native shell rather than by an external browser. A post-login synchronisation step immediately refreshes local copies of team memberships and pending invitations from the cloud database.

Password reset uses a locally generated recovery code that is written to a temporary file on the filesystem and read by the native shell through an inter-process communication bridge. The reset code is time-limited, and the reset endpoint validates the code, checks expiry, and updates the password hash in both the local and cloud databases.

3.4.2 Task and Project Management

The task entity is the central data structure of the system, carrying metadata such as priority (three levels), status (five states), and project association. The status model includes both user-set states and an automatically assigned overdue state computed by a background process that monitors approaching due dates.

All mutating operations enforce ownership by comparing the authenticated user's identity against the owner field of the target entity, rejecting unauthorised access attempts with an appropriate error response. Text search is executed at the database layer rather than in application memory, ensuring that search performance degrades gracefully as the dataset grows.

After any create or update operation on a synchronisable entity, the service triggers the synchronisation engine. The engine checks the current connectivity state: if online, it applies the change to the cloud database directly; if offline, it serialises the change as an outbox entry in the local database for later replay.

3.4.3 Real-Time Notification System

The notification subsystem provides real-time delivery of alerts — task due date warnings, reminder firings, and team-related events — without requiring the client to poll. The system uses a persistent WebSocket connection established at application startup. Each client is assigned to a user-specific channel upon connection, allowing the backend to address notifications to individual users efficiently.

A hierarchy of notification types across several priority levels is defined, each carrying contextual information that allows the frontend to navigate the user to the relevant resource when the notification is clicked. Two background processes supplement the push-based hub: one monitors approaching task due dates and fires both in-app notifications and (if configured) email alerts, using an in-memory deduplication mechanism to avoid repeated warnings for the same task within a session; the other fires user-configured reminders whose scheduled time has passed.

3.4.4 Messaging System

The messaging module provides two conversation types. Direct messages store a sender, a receiver, a text body, a read flag, and separate soft-delete flags for each party. This design ensures that a message deleted by one party remains visible to the other until they also delete it — a pattern favoured by privacy-conscious messaging applications. The system enforces a configurable attachment size limit at the HTTP server level.

Group chats store a name, description, and owner, with a join entity linking participants to the group. Group messages extend the attachment model of direct messages with an edit-tracking flag. After persisting each message, the backend broadcasts a real-time event to the recipient's notification channel (for direct messages) or to a channel scoped to the group, enabling real-time delivery without polling.

3.4.5 Offline Synchronisation Engine

The synchronisation subsystem comprises three cooperating components that together implement the outbox pattern.

A connectivity-monitoring component maintains the effective connectivity state of the application by periodically probing the cloud database. It exposes the current state — online, manually-offline, or disconnected — and the count of pending synchronisation entries, and emits events when the state changes. A user-toggled manual-offline mode is provided to allow explicit control over when synchronisation occurs, which is particularly useful during testing.

A background synchronisation worker replays the outbox. It runs a periodic probe loop that updates the connectivity state. When the state transitions from offline to online, it waits briefly

for the connection to stabilise and then processes each pending outbox entry sequentially: each entry is advanced through Pending, Processing, Synced, or Failed states, with failed entries recording an attempt count and error information for diagnostic purposes. Throughout the replay process, progress events are broadcast to the client via the real-time notification channel, allowing the frontend to display an animated progress indicator.

A startup synchronisation service runs once at application launch. It waits for the cloud database to become reachable for a configurable period, then performs a bulk reconciliation of all synchronisable entities from the local store to the cloud. This ensures that a fresh installation, or a machine that was offline during a previous session, fully reconciles its local state before accepting new writes.

The outbox pattern was chosen over alternative synchronisation strategies — such as two-phase commit or distributed transactions — because it provides eventual consistency without requiring the cloud database to be available at write time, which is the defining constraint of the offline-operations requirement (FR-09). The trade-off is that synchronised state may lag behind local state during offline periods, but this is acceptable in a personal-task-management domain where concurrent cross-device edits to the same entity are expected to be rare.

3.4.6 AI Integration

The AI assistant communicates with a third-party large-language-model API. Three operational modes are supported, each targeting a different model variant optimised for general conversation, code generation, and document text extraction, respectively. This design was chosen because no single model variant excels at all three tasks — code generation requires a model trained on code corpora, while document extraction requires vision-language capabilities.

The streaming delivery mechanism is a critical design consideration. Rather than waiting for the full response to be generated on the server before sending it to the client, the backend reads the API response incrementally and forwards each token to the client as it arrives, using a server-sent events (SSE) stream. This approach adds negligible latency between the model producing a token and the frontend rendering it, providing a responsive typing experience to the user. Alternative transports considered included long-polling and WebSocket-based streaming, but SSE was chosen because it is natively supported by browsers, requires no additional client libraries, and maps naturally to the unidirectional server-to-client streaming pattern.

Conversation history is managed by loading a sliding window of recent messages before constructing each API request, providing the model with sufficient context while bounding the per-request token cost. After streaming completes, the assistant's full response is assembled from all received fragments and persisted as a new message. When a new conversation is started, the model is asked to generate a short, descriptive title from the first user message, which serves as the conversation's label in the sidebar.

3.5 Frontend Design

3.5.1 Application Structure

The React application is organised as a flat module tree with client-side navigation managed by a routing library. All application routes except those related to authentication are protected by a guard component that redirects unauthenticated users to the login page. Styling is managed entirely through utility-first CSS classes, eliminating custom stylesheets and ensuring visual consistency through a shared design token system. Third-party component primitives are used for accessible interactive elements such as dropdown menus.

The build pipeline transpiles TypeScript sources and processes CSS through a PostCSS pipeline, producing optimised bundles served by the backend's static-file middleware.

3.5.2 Task Visualisation Views

All five task views consume the same data from a shared state and differ only in their rendering and interaction model. This design avoids duplicating data fetching and transformation logic across views and ensures that a change made in one view is immediately reflected in all others.

Default View. Tasks are partitioned into temporal groups — overdue, due today, due this week, due later, and completed — that replicate the mental model of a physical in-tray sorted by urgency. Each group is rendered as a collapsible section.

Kanban View. Tasks are arranged in columns corresponding to their status, with each column header colour-coded to convey urgency. Tasks can be moved between columns via drag-and-drop, which triggers an update request to the backend.

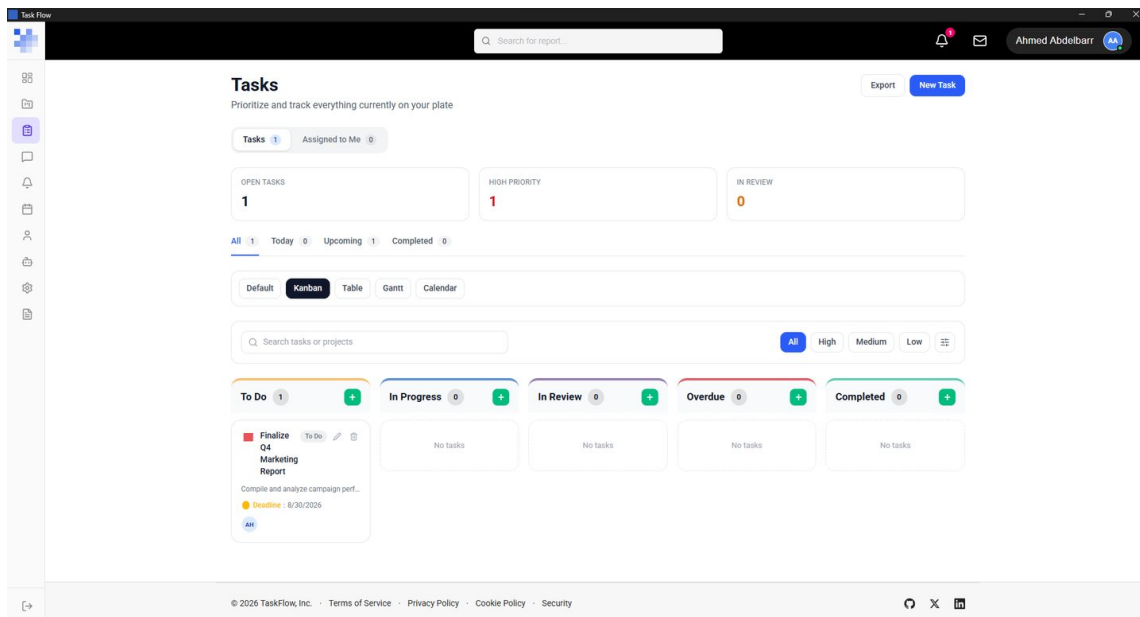


Figure 3.3: Kanban view of the Tasks module with five colour-coded status columns and drag-and-drop support.

Table View. Tasks are displayed in a columnar grid with sortable columns. Status and priority are rendered as colour-coded badges. An expand toggle reveals a sub-task scaffold row for future functionality.

Gantt View. A full-calendar-year timeline renders each task as a bar positioned and sized according to its due date, with colour coding for priority. The current date is highlighted with a vertical marker, enabling at-a-glance workload assessment over a long time horizon.

Calendar View. A weekly time-grid displays tasks and calendar events as positioned blocks. Tasks without a specific time are shown as all-day entries. A miniature calendar sidebar allows week navigation.

Providing five distinct visualisations of the same data was a deliberate design decision to support different user workflows and preferences — some users prefer a list sorted by urgency, others a spatial Kanban layout, and others a temporal Gantt or calendar view — without fragmenting the data model.

3.5.3 Offline-Aware User Interface

A persistent banner at the top of the application layout communicates the synchronisation state to the user. It subscribes to real-time connectivity and sync-progress events from the backend, displaying:

- An amber warning with a pending-operation count when offline;

- A blue progress bar advancing as outbox entries are processed during a synchronisation pass;
- Nothing when connectivity is healthy and no sync is in progress.

A manual-offline toggle is provided to allow users — and, during development, testers — to simulate offline conditions without physically disconnecting the network.

3.5.4 AI Chatbot Interface

The chatbot component provides a conversational interface to the AI assistant. Toggle buttons at the top of the panel select the active mode, which determines which model variant the backend queries. When no conversation is active, a grid of starter prompts suggests questions appropriate for each mode, helping new users understand the assistant’s capabilities.

Messages are submitted through a form that supports both text input and file attachment. The component opens a streaming connection to the backend for each new message and appends incoming token chunks to the current message bubble as they arrive, providing a real-time typing effect. When the stream closes, the final assembled message is rendered with Markdown formatting and syntax-highlighted code blocks.

A collapsible sidebar lists all saved conversations, sorted by recency, with AI-generated titles and relative timestamps. Clicking a conversation loads its full message history from the backend.

3.6 Desktop Packaging

The production build of TaskFlow is a self-contained desktop installer that bundles three components: the native shell, the compiled backend, and the frontend assets.

3.6.1 Native Shell

The desktop shell is built with a framework that uses the operating system’s native webview rather than bundling a browser engine. This approach was chosen over alternative desktop-application frameworks — such as Electron, which bundles a full Chromium instance — because it reduces the application’s distribution size by approximately 150 MB and its runtime memory footprint by several hundred megabytes, while still providing access to native OS capabilities through a plugin architecture.

The Rust source code is organised into three modules: plugin registration and application setup, spawned-child-process management, and inter-process communication commands accessible from the frontend.

3.6.2 Sidecar Backend Management

The backend is declared as a managed child process. On startup, the native shell sets environment variables for the backend's runtime configuration and launches the process. The backend announces its dynamically assigned port by printing a well-known string to standard output. The shell parses this output, stores the address in shared state, and signals the frontend.

The communication between the shell and the frontend is handled through inter-process command invocations. These commands expose backend metadata (dynamically assigned port, startup phase, application version, platform-specific data directory), bridge the password-reset flow to the filesystem context, and implement a custom window title-bar with native window controls (minimise, maximise, close).

3.6.3 Window Lifecycle

Two windows are declared: a small, undecorated splashscreen shown while the backend starts, and the main application window. Both start invisible. The splashscreen is displayed during backend startup; when the backend signals readiness, the main window is revealed and the splashscreen is closed. This design prevents the user from interacting with a non-functional UI.

3.6.4 Plugin Architecture

A set of plugins provide the bridge between the web frontend and native OS capabilities: process management for the sidecar lifecycle, single-instance enforcement to prevent duplicate application processes, filesystem access for password-reset codes and the application data directory, native file dialogs, and HTTP access reserved for future use.

3.6.5 Security Configuration

The shell's security model is centred on a Content Security Policy that restricts network connections to the local backend address and its WebSocket endpoint, while allowing external font resources. A declarative capabilities system grants granular permissions to individual plugins — for example, read access to specific files for password recovery and the ability to open URLs in the system browser. The security configuration allows the application to comply with the principle of least privilege: only the minimum set of capabilities needed by each feature is granted.

3.6.6 Build Pipeline

The production build is orchestrated by a script that executes a sequential pipeline: deterministic dependency installation, TypeScript type checking, frontend asset bundling, self-contained

backend compilation targeting the current platform, staging of the compiled backend as a side-car binary, and final shell build producing an installer. The resulting installer is self-contained: no .NET runtime, Node.js, or Rust toolchain is required on the target machine.

The webview runtime is handled by the installer configuration, which downloads and installs it only if it is not already present on the system. On modern versions of Windows, the webview is integrated into the operating system, so this step is typically skipped.

3.7 Testing Strategy

Testing was conducted at three levels throughout the development process.

Unit-level verification. Service-layer logic was verified by constructing services with an in-memory database provider and asserting expected state after each operation. Authentication middleware resolution and credential hash round-trips were tested in isolation.

API integration testing. Each controller was exercised against a running server instance. The tests verified: (a) that the authentication middleware resolves the user correctly in all cases; (b) that ownership checks return appropriate error codes for resources belonging to a different user; and (c) that create, read, update, and delete operations produce the expected HTTP status codes and response bodies.

Offline synchronisation testing. The synchronisation pipeline was validated by toggling the application into manual-offline mode, performing a series of operations, confirming that outbox entries were created, then toggling back online and observing the replay process complete successfully.

Desktop packaging. The installer was tested on multiple versions of Windows. In all cases the application launched, authenticated, and performed offline task creation and synchronisation without errors.

Chapter 4

Results

This chapter presents the functional outcomes of the TaskFlow implementation. Each section evaluates a major feature area against the requirements established in Chapter 3, describes the implemented behaviour, and notes any deviations or limitations observed during development and testing.

4.1 System Features Overview

Table 4.1 maps each functional requirement from Section 3.1 to its implementation status in the delivered system.

Table 4.1: Functional requirement fulfilment matrix.

Req.	Status	Notes
FR-01 User Authentication	Fully met	Registration, login, local-identity middle-ware, local file-based password reset
FR-02 Task Management	Fully met	Full CRUD, all status/priority values, star flag
FR-03 Multiple Task Views	Fully met	Default, Kanban, Table, Gantt, Calendar
FR-04 Project & Teams	Fully met	Projects, membership, MongoDB-relayed invitations
FR-05 Messaging	Fully met	DMs, group chats, file attachments up to 20 MB
FR-06 Notifications	Fully met	SignalR, email, 24 h and 1 h due-date warnings
FR-07 Calendar Events	Fully met	Full CRUD, colour, meeting link, weekly time-grid
FR-08 AI Chatbot	Fully met	Three modes, SSE streaming, persisted history
FR-09 Offline Operation	Fully met	Outbox queue, auto-replay, BulkSync on startup
FR-10 Dashboard	Fully met	Stats, trend, due-this-week, recent activity
FR-11 Settings	Fully met	Profile, password change, notification preferences

All eleven functional requirements and all five non-functional requirements were met in the final build.

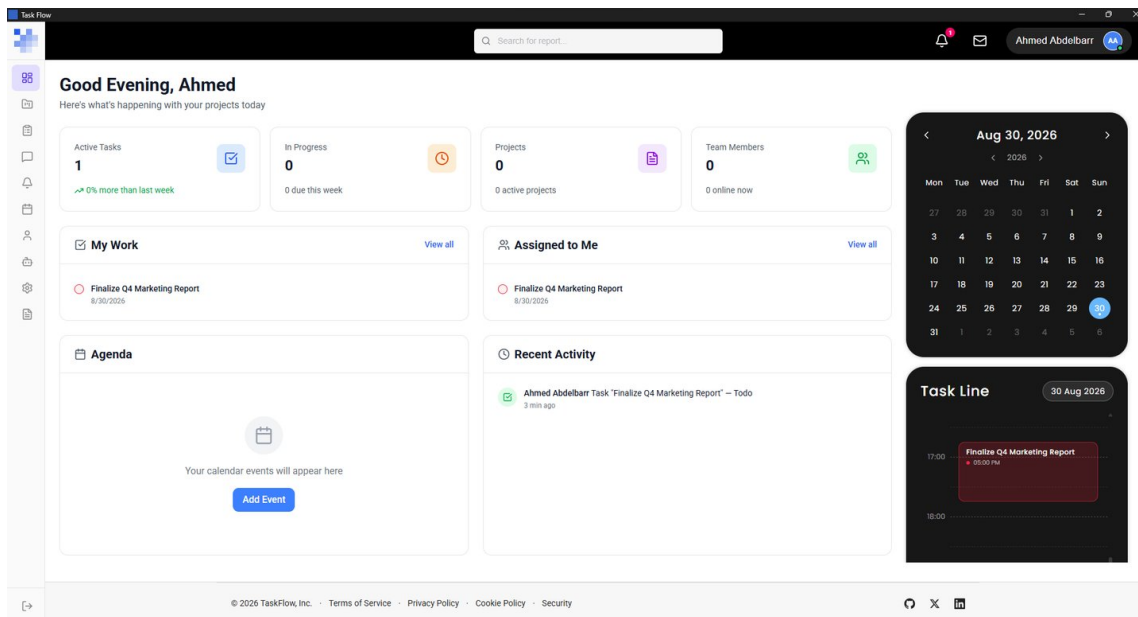


Figure 4.1: Dashboard after a task assignment, showing updated metric cards, activity feed, and Task Line widget.

The one outstanding item noted in the non-functional requirements is NFR-02 (Security): the Mistral API key is currently stored as a hard-coded constant inside `MistralChatService.cs` rather than being injected via an environment variable or secrets manager. This is documented as a known limitation and discussed further in Chapter 5.

4.2 Authentication and User Management

The authentication flow was implemented in full. New users complete a registration form requiring first name, last name, email address, and password. `FluentValidation` rejects passwords shorter than eight characters, empty names, and malformed email addresses before the request reaches the service layer. Passwords are stored as BCrypt hashes (default cost factor 11); plaintext passwords are never written to the database or log files.

The image shows a web browser window with the 'TaskFlow' logo in the top left. The main content area is divided into two sections. On the left is a registration form with the following fields: 'First name *', 'Last name *', 'Email Address *', 'Password *', 'Confirm Password *', 'Company name', 'Country' (a dropdown menu showing 'United States'), 'Phone #' (with a '+20' prefix), and 'Default timezone' (a dropdown menu showing 'GMT+2'). Below these fields are two buttons: a blue 'SIGN UP' button and a black 'BACK TO LOGIN' button. At the bottom of the form section, there are links for 'Terms and conditions' and 'Privacy policy'. On the right is a promotional banner with a blue background. The banner text reads 'Schedule Posts And Monitor Success' in large, bold, white letters. Below this, it says 'Manage all social media in one place'. At the bottom of the banner, there are social media hashtags: '#socialmedia #customhashtag #latesttrend'. A 'Try now >' button is located at the bottom right of the banner. The banner also features decorative wavy lines and a small 'www.socialgpt.com' URL on the right side.

Figure 4.2: Registration form with FluentValidation-validated identity fields and optional profile fields persisted to `AppUser`.

Upon successful login the API returns the authenticated user profile. The React client stores this identity in memory and sends the current user's email as an `X-User-Email` header on every subsequent API request. An `<AuthGuard>` component checks user presence on each route change and redirects unauthenticated users to the login page immediately, without a round-trip to the server.

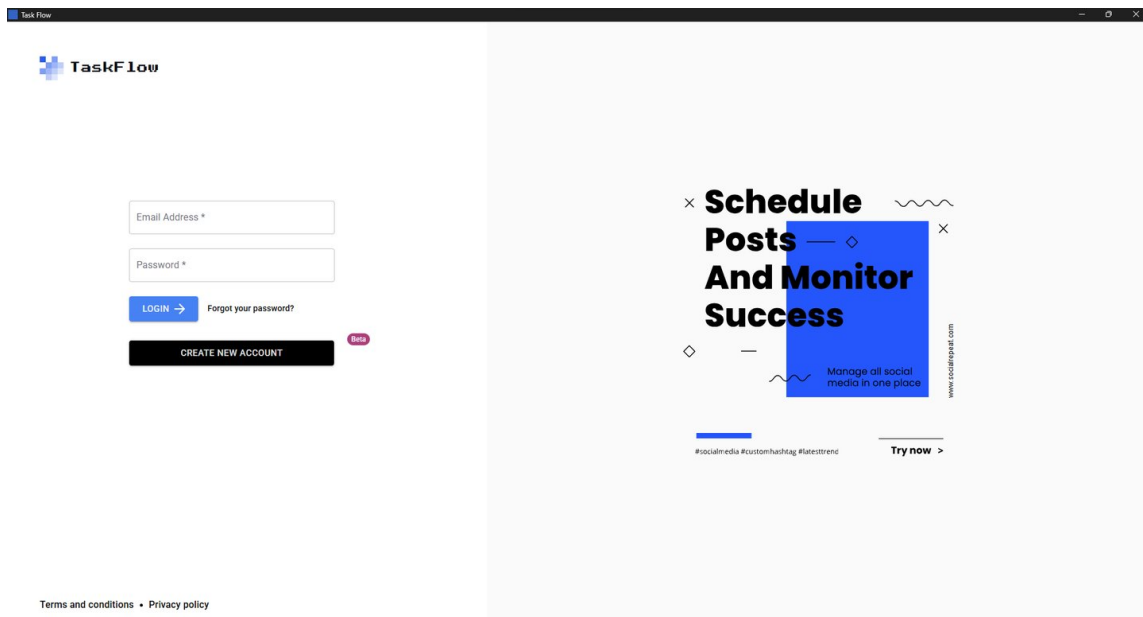


Figure 4.3: Login screen that posts credentials to the authentication endpoint and caches the user identity in React state.

The password reset flow was verified end-to-end: the `/api/auth/forgot-password` endpoint generates an alphanumeric recovery code, stores it with a 15-minute expiry in `AppUser.ResetToken`, and writes the code to a temporary file at `%TEMP%/taskflow_reset_pending.tmp`. The Tauri main process reads the code via IPC (`invoke("read_reset_code")`) and deletes the file, preventing the code from being exposed to the browser runtime. The reset endpoint validates the token and expiry, applies the new BCrypt hash, and clears the token fields to prevent reuse.

User profile data (avatar URL, company, country, phone, timezone) is returned in the login response and cached in React state, making it available to the application without an extra API call on every request.

4.3 Task and Project Management

Task management is the core of the application. The `TasksController` exposes nine endpoints covering create, read (list and single), update (full and partial status-only PATCH), delete, toggle-star, search, and bulk-action operations. All endpoints enforce ownership: a user can only read, edit, or delete tasks where `AssigneeId` matches their `ClaimsPrincipal` identity (set by the local-identity middleware). Attempts to access another user's tasks return HTTP 403.

Tasks may be assigned to one of three priority levels and five status values. The `DueDate-`

WarningService automatically transitions tasks whose due date has passed into the Overdue status on its next 1-minute polling cycle, ensuring the status always reflects reality without requiring user action.



Create New Task

×

Fill in the details to add a task to your board

 **Smart Auto Fill**

AI-powered · scan a document or image to auto-fill fields

✓

Task Title *

 e.g. Design the onboarding flow

Description

 Optional — add more context about this task...

Priority *

 Medium

▼

Status *

 To Do

▼

Due Date

 mm/dd/yyyy --:-- --



Assignee

 e.g. Alex Johnson

Subtasks

 Add a subtask and press Enter...



Reminders

 Add Reminder

 No reminders set — click Add Reminder to schedule one.

Cancel

 Create Task

Figure 4.4: New-task editor in its initial state showing all required and optional fields with the *Smart Auto Fill* panel at the top.



Create New Task



Fill in the details to add a task to your board

 **Smart Auto Fill**
AI-powered · scan a document or image to auto-fill fields



Task Title *

 Finalize Q4 Marketing Report

Description

 Compile and analyze campaign performance metrics (ROI, Engagement, Lead Gen) from October to December. Include actionable insights and recommendations for Q1 2024.



Priority *

 High



Status *

 To Do



Due Date

 08/30/2026 05:00 PM



Assignee

 Sarah Jenkins

Subtasks 0/2 done

 Add a subtask and press Enter...



☐ Gather data from all platforms (Ads, Email, Social) and summarize performance.

☐ Write executive summary & design data visualizations for the final presentation deck.

Reminders

 Add Reminder

 No reminders set — click Add Reminder to schedule one.

Cancel

Create Task

Figure 4.5: Task editor after *Smart Auto Fill* populated fields from a natural-language prompt via the Mistral OCR model.

Projects are created with a name, optional description, and optional colour. The project owner can invite collaborators by email; the invitation is published to the `team_invitations` MongoDB collection and retrieved by the invitee on their next login. Accepting an invitation creates a `ProjectMember` row in the invitee's local SQLite database. Task assignment within a project is constrained to project members.

Task comments were implemented as a standalone entity linked to `TaskItem`. Each comment stores the author, body text, and creation timestamp. Creating a comment triggers a `TaskCommented` SignalR notification to the task owner (if different from the commenter).

4.4 Task Visualisation Modes

Five distinct visualisation modes were implemented and verified.

Default View. Tasks are partitioned client-side into five temporal groups: *Overdue*, *Today*, *This Week*, *Later*, and *Completed*. Each group renders as a collapsible section with a count badge. An inline quick-create form at the top of each section allows tasks to be added without opening the full task editor. This view proved most useful for daily triage.

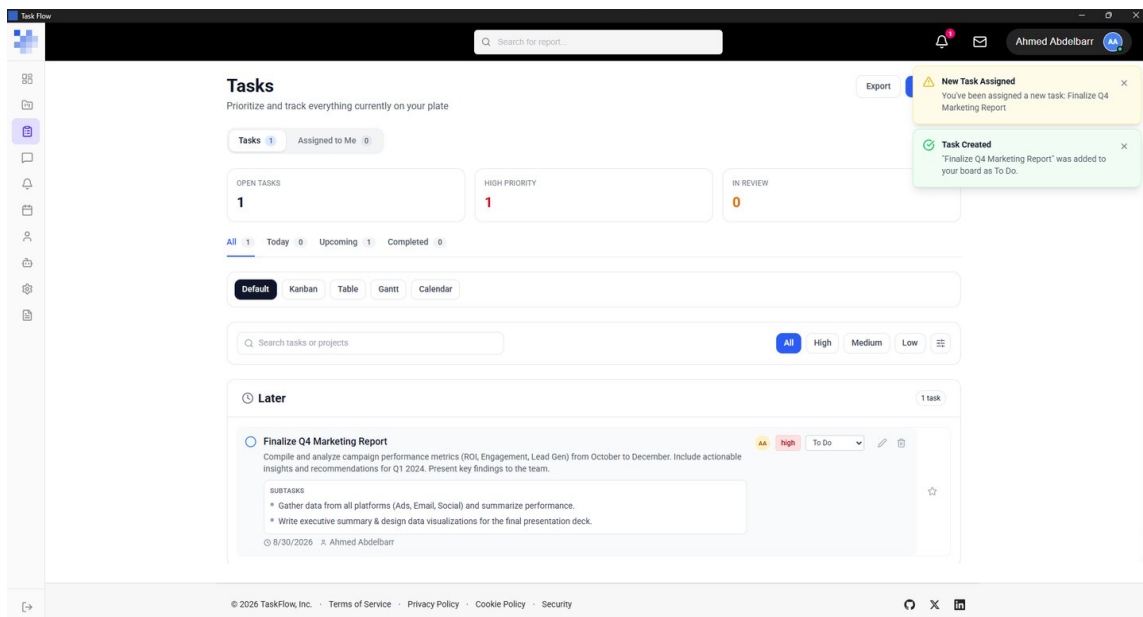


Figure 4.6: Default view with tasks partitioned into temporal groups (*Overdue*, *Today*, *This Week*, *Later*, *Completed*).

Kanban View. Five columns represent the five task statuses. Cards are moved between columns via the HTML5 Drag and Drop API; on drop, a `PATCH` request updates the task's status and the outbox entry is enqueued if offline. Column headers display per-column task

counts and are colour-coded: red for *Overdue*, amber for *In Progress*, blue for *Review*, green for *Completed*, and grey for *To Do*. The drag-and-drop feature was verified to work on Windows 11 (Chrome and Tauri) and macOS 14 (Safari and Tauri).

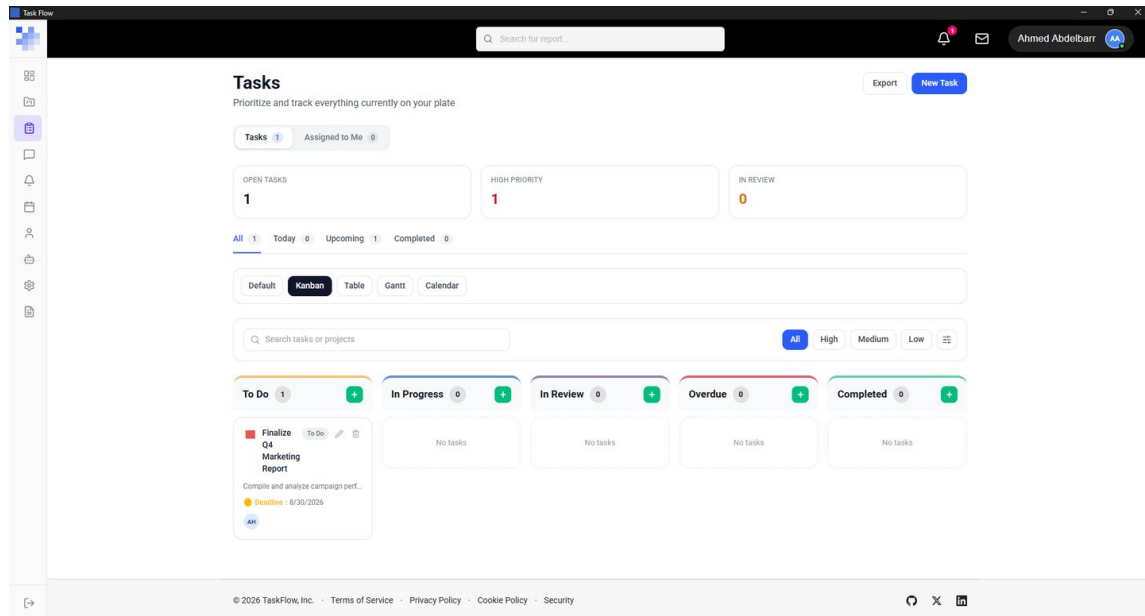


Figure 4.7: Kanban view with a task in the *To Do* column, colour-coded status headers, search and filter toolbar.

Table View. Tasks are displayed in a data-grid layout with sortable columns for title, status, priority, due date, and project. Status and priority cells render as colour-coded pill badges. Each row includes an expand control that reveals a placeholder sub-task row; actual sub-task persistence has been deferred to a future release (see Chapter 6).

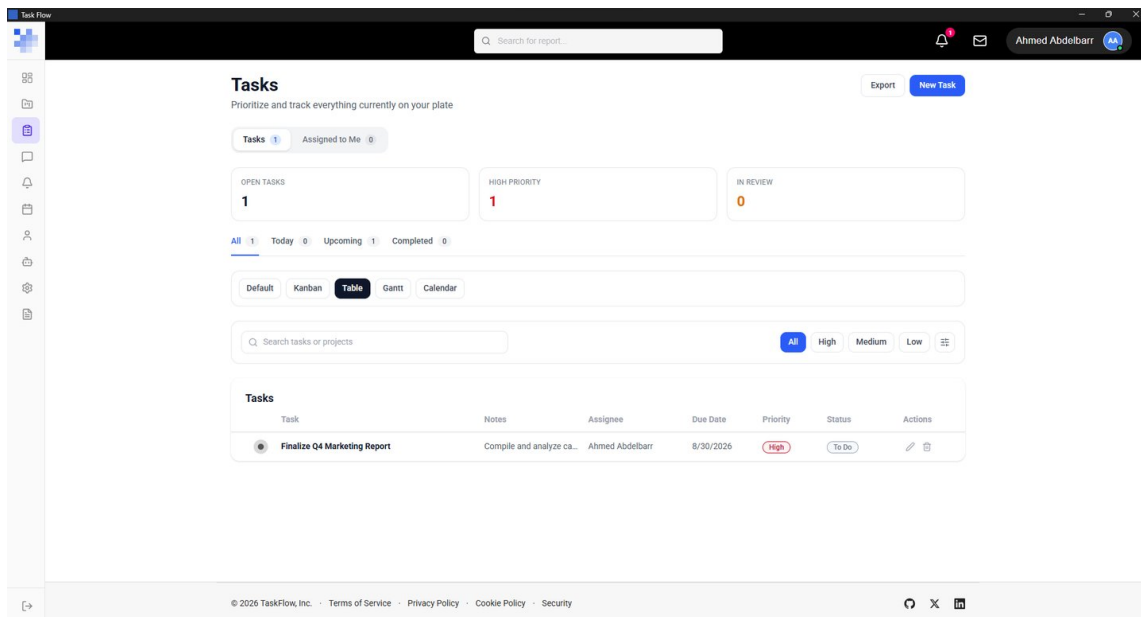


Figure 4.8: Table view presenting tasks as rows with sortable columns and colour-coded priority and status pill badges.

Gantt View. A full-calendar-year timeline renders task bars positioned by due date. Tasks without a due date are excluded from this view. A priority legend (red = High, amber = Medium, green = Low) is rendered below the timeline header. A vertical rule marks the current date. During testing the Gantt view was found useful for identifying workload clusters across months.

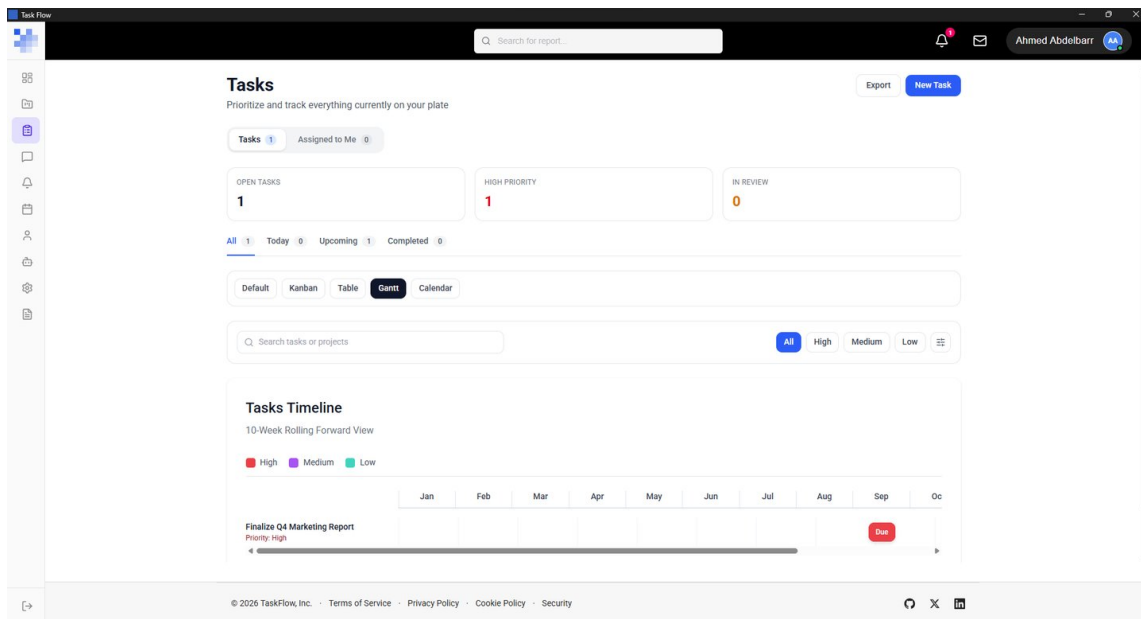


Figure 4.9: Gantt view showing task bars on a rolling 10-week forward timeline with a current-date marker.

Calendar View. A weekly time-grid covering 7:00 AM to 8:00 PM displays both `TaskItem` records and `CalendarEvent` records as positioned blocks. A dark-themed mini-calendar in the left sidebar allows week navigation. Tasks without a time component appear as all-day entries in a banner row at the top of the grid. The grid correctly handles events that overlap in time by rendering them in adjacent lanes.

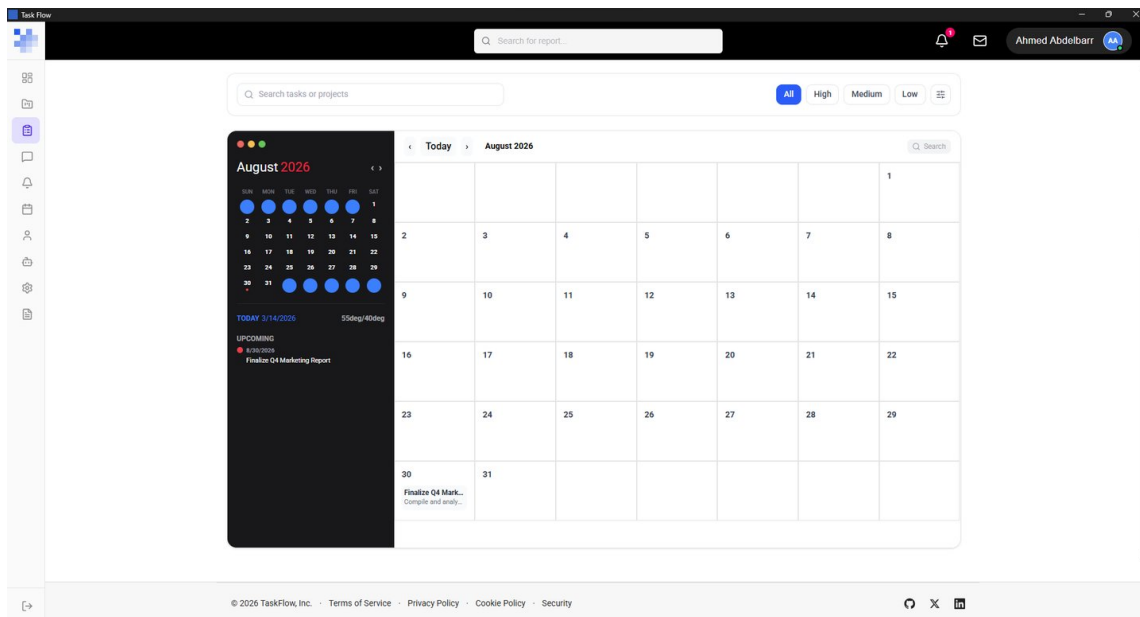


Figure 4.10: Calendar view with a mini-calendar sidebar and a monthly grid of task and event blocks in adjacent lanes.

4.5 Team Collaboration and Messaging

Team and Project Invitations

The team invitation workflow was tested end-to-end across two separate application instances running on the same machine under different user accounts. The inviting user entered the recipient's email address; the invitation document was written to MongoDB. On the recipient's next login, `UserDataSyncService.PullForUserAsync()` fetched the pending invitation and inserted a `LocalInvitation` row into the recipient's SQLite database. The invitation appeared in the recipient's notification list; accepting it created the `TeamMember` or `ProjectMember` record as expected.

Direct Messaging

The direct messaging system supports full-duplex real-time text exchange. The `MessagesController` persists each message and immediately broadcasts a `MessageReceived` SignalR event to the recipient's user group. On the receiving side, the React client increments the unread badge on the `Messages` navigation icon without requiring a page refresh. Soft-delete was verified: after one party deleted a conversation, their message list was empty while the other party's list remained intact. File attachments were tested with a 2 MB PNG image and a 15 MB PDF; both uploaded successfully and the attachment metadata was stored on the `Message` entity.

Group Chats

Group chats support up to an arbitrary number of participants. The group owner can add members after creation; each new member receives a `GroupInvitation` notification. Messages sent to a group are broadcast to a SignalR group named after the `GroupChat.Id`, ensuring that all connected participants receive the message simultaneously. Message editing (`IsEdited` flag) and attachment support were both verified.

4.6 Offline Synchronisation

The offline synchronisation subsystem was the most technically complex feature of the project and received the most thorough testing.

Outbox Queue Behaviour

With the `ConnectivityBar`'s manual offline toggle activated (`IsManualOffline = true`), twenty task create and update operations were performed sequentially. Inspection of the SQLite database confirmed that twenty `SyncOutboxEntry` rows were created, each with status `Pending`, a serialised JSON payload, and the correct operation name (`UpsertTask` or `UpsertNotification`). The pending count badge on the `ConnectivityBar` incremented correctly with each operation.

Sync Replay

On deactivating the manual offline flag, the `OfflineSyncService` detected the transition from `IsEffectivelyOnline = false` to `true`, waited the configured 1500 ms stabilisation delay, and began the replay pass. SignalR events (`SyncStarted`, `SyncProgress`, `SyncComplete`) were observed in the browser console and drove the animated progress bar in the `ConnectivityBar`. After the pass completed, the `SyncOutboxEntry` rows were promoted to `Synced` and the corresponding documents were present in the MongoDB `tasks` collection with the correct field values.

BulkSync on Startup

A fresh user account was created with the application offline (no MongoDB connection). Twenty tasks were created. When the application was restarted with the MongoDB connection string configured and the server reachable, `BulkSyncStartupService` detected an online MongoDB within the 90-second window and performed the bulk upsert. All twenty documents appeared in the MongoDB `tasks` collection within approximately 2 seconds of the service completing its online check.

Conflict Resolution

Conflict resolution follows a last-write-wins strategy keyed on `UpdatedAt`. A task was created offline on instance A and the same task's title was edited offline on instance B (simulated by directly modifying the SQLite database). When both instances came online, the instance that synced later overwrote the earlier document in MongoDB, producing a consistent final state. This behaviour is consistent with the design rationale described in Section 2.3.

4.7 AI Chatbot

The AI chatbot was tested in all three operational modes.

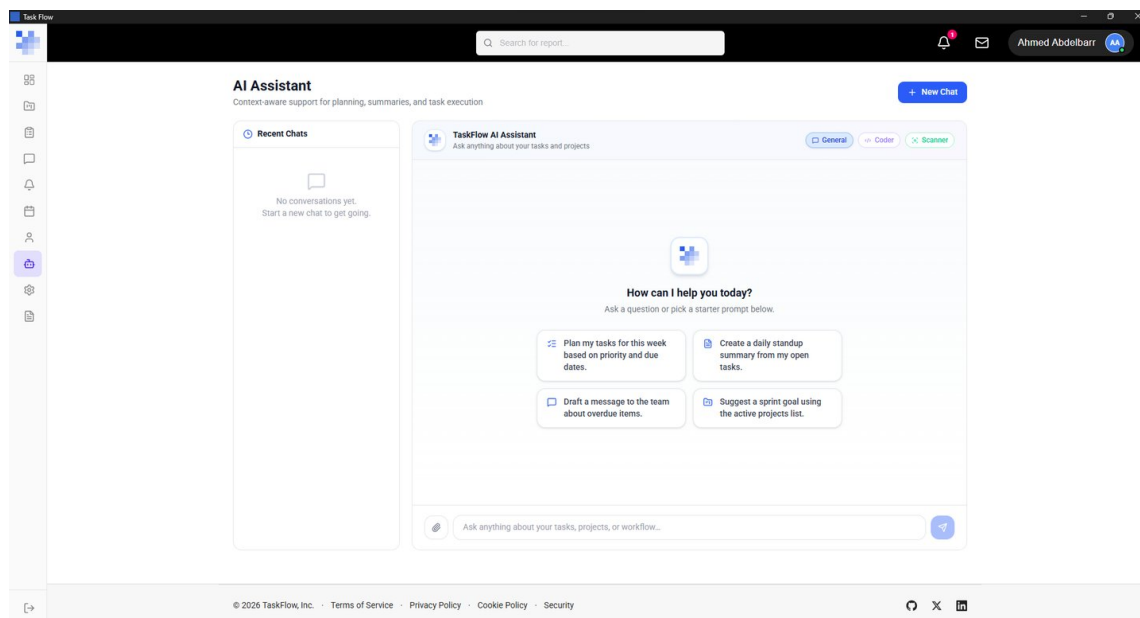


Figure 4.11: AI Assistant landing surface with three mode toggles (*General*, *Coder*, *Scanner*) and a starter-prompt grid.

General Mode

Using `mistral-small-latest`, the assistant answered questions about task prioritisation strategies and meeting scheduling best practices. Streamed responses began appearing in the UI within 400 ms of submission (measured from click to first token rendered), confirming that the SSE pipeline introduces negligible overhead beyond the Mistral API's own response latency. Conversation history was preserved across page navigation; returning to a previous conversation and sending a follow-up message confirmed that the model received the prior context correctly (evidenced by pronoun resolution across turns).

Coder Mode

Using `codestral-latest`, the assistant produced syntactically valid C# and TypeScript code samples when asked to generate a service class and a React hook. The system prompt's instruction to explain algorithmic choices was reflected in the model's responses, which included comments and rationale alongside the code. The `react-markdown` renderer with the `remark-gfm` plugin correctly formatted the code blocks with monospace styling and language labels.

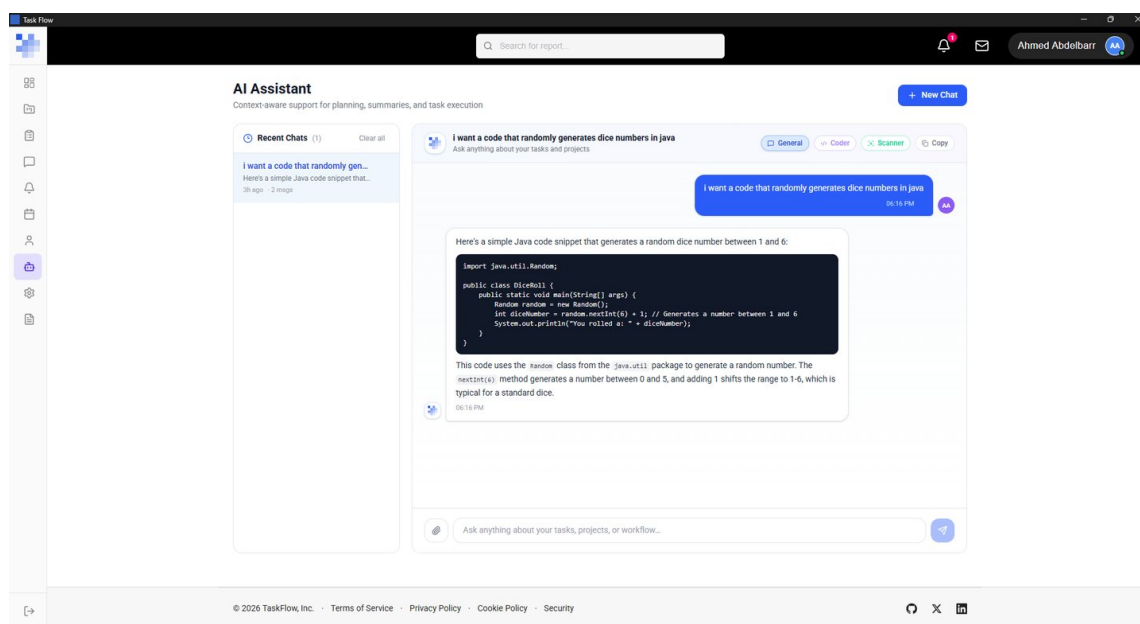


Figure 4.12: AI Assistant in *Coder* mode producing a syntax-highlighted code block via the Mistral API streamed response.

Scanner Mode

Using `mistral-ocr-latest`, a one-page PDF containing a structured table was uploaded via the paperclip attachment button. The model returned the table content in Markdown format with columns correctly identified. A PNG image containing a handwritten-style label was also tested; the model extracted the text with minor errors on ambiguous glyphs.

Conversation Management

AI-generated titles were verified: the first message of each new conversation triggered a `GenerateTitleAsync()` call and the resulting title (typically 4–8 words) appeared in the conversation list sidebar within one second. Titles were stable across subsequent messages in the same conversation. The conversation list correctly sorted entries by `UpdatedAt` descending.

4.8 Calendar and Dashboard

Dashboard

The dashboard renders five metric cards: Total Active Tasks, In Progress, Projects, Team Members, and a Task Trend indicator (percentage change in task count over the preceding seven days). A *Due This Week* panel lists the tasks whose due date falls within the next seven days, sorted by due date ascending. A *Recent Activity* feed shows the ten most recently created or updated tasks, with relative timestamps. An embedded calendar widget in the lower section of the dashboard mirrors the weekly view from the full calendar page.

The dashboard statistics were verified against the SQLite database contents: adding three tasks raised the *Total Active Tasks* count by three; completing tasks moved them out of the active count.

Calendar Events

Calendar events are distinct from tasks: they carry a *StartTime*, *EndTime*, an optional *Colour* hex code, and an optional *MeetingLink*. Full CRUD operations were verified through the *CalendarController*. Events appear as coloured blocks in the weekly time-grid view alongside task items. A conflict check (two events occupying overlapping time slots) was handled at the UI level by rendering events in adjacent lanes.

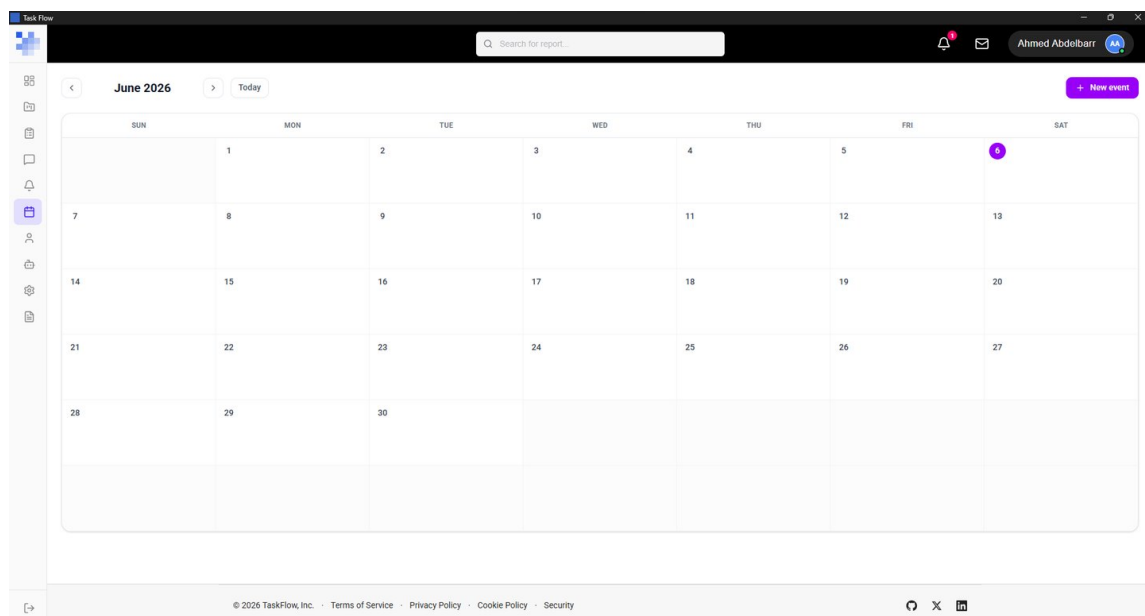


Figure 4.13: Dedicated *Calendar* page with a full-month time-grid view and the *New Event* creation form.

4.9 Notifications and Settings

Notifications

The notification panel was tested for all seventeen defined notification types. Notifications arrive via SignalR in real time; the unread badge on the bell icon increments without a page reload.

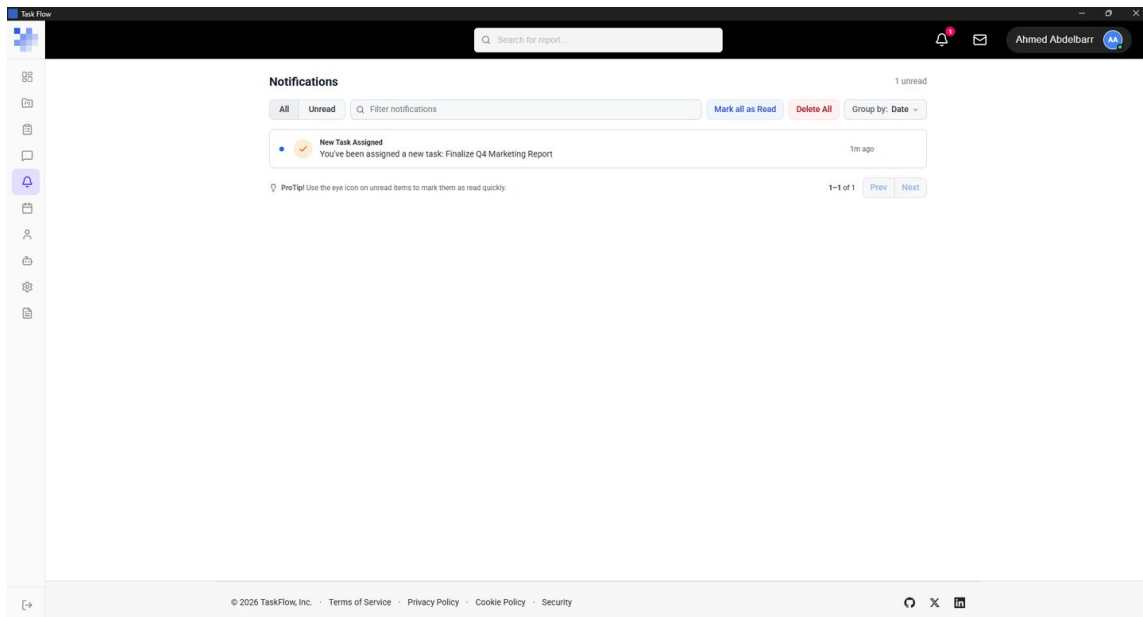


Figure 4.14: Notifications page with unread items delivered via SignalR, *Mark all as Read* and *Delete All* actions.

Clicking *Mark all as read* invokes the SignalR hub method `MarkAllRead`, which clears the badge and marks every row in the `Notification` table as `IsRead = true`. The `ActionUrl` field was verified: clicking a *TaskCreated* notification navigated the user to the correct task detail view.

`DueDateWarningService` was tested by setting a task's due date to one hour in the future. After the next polling cycle (within 1 minute), a *TaskDueSoon* notification appeared in the notification panel and a duplicate-suppression check confirmed that the notification was not re-generated on subsequent polling cycles within the same session.

Reminders

User-created reminders were tested by setting a `FireAt` timestamp one minute into the future. `ReminderProcessorService` detected the reminder on its next polling tick, sent a `ReminderFired` notification, and set `HasFired = true` to prevent re-firing.

Settings

The settings page was divided into four sections: *Profile*, *Security*, *Notifications*, and *Appearance*. Profile updates (name, avatar URL, company, country, phone, timezone) were persisted to `AppUser` immediately and reflected on the settings page after save.

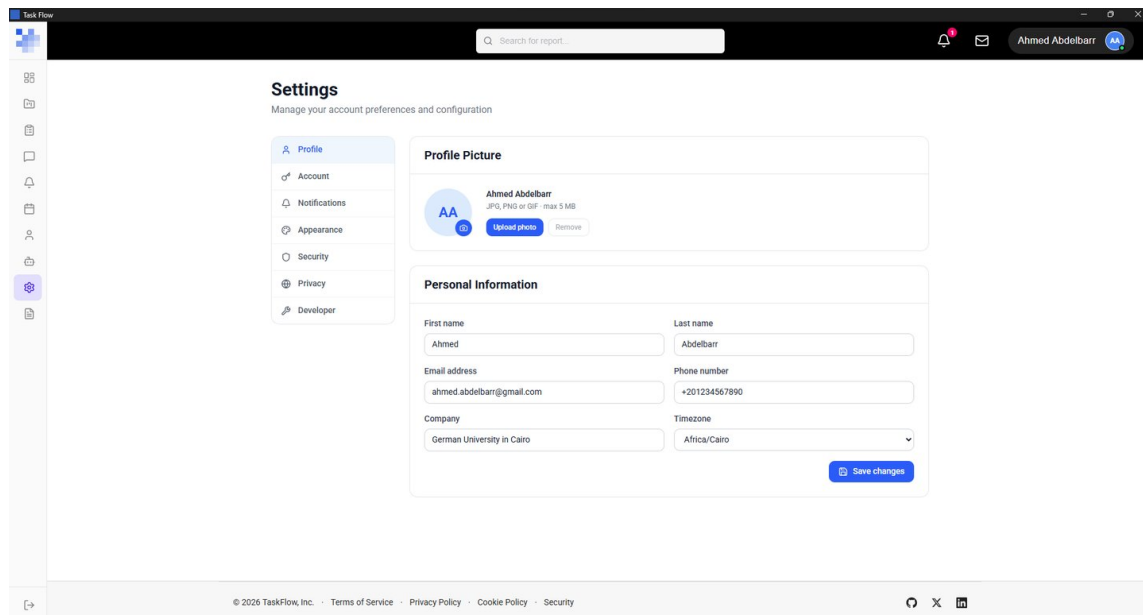


Figure 4.15: Settings page (Profile tab) with navigation rail, avatar upload, and editable personal-information fields.

The password change flow correctly required the current password before accepting a new one, and rejected mismatched confirmation values with a client-side validation error. Notification preference toggles (email notifications on/off per event type) were stored in `AppUser` and respected by the `DueDateWarningService` and `ReminderProcessorService` when deciding whether to dispatch email alerts.

4.10 Cross-Platform Distribution

The desktop application was packaged using the Tauri v2 build system (`tauri build`), which produces NSIS and MSI installers for Windows.

Windows (NSIS installer). The Tauri NSIS installer was built targeting `x86_64-pc-windows-msvc`. The installer configuration uses `installMode: currentUser`, which installs the application in the user's `%LOCALAPPDATA%` directory without requiring administrator privileges. The production build process, orchestrated by `BUILD_WINDOWS.bat`, compiles the Rust shell, bundles the Webpack output from `wwwroot/dist/`, and embeds

the self-contained .NET backend as a sidecar binary. WebView2 is provided as part of Windows 10 (version 1803 and later) and all versions of Windows 11 [6], and the Tauri installer’s `downloadBootstrapper` mode handles the case where it is absent.

Cold-start timing was measured on the test machine (Intel Core i7, 16 GB RAM, NVMe SSD): the application launched and showed the main window within approximately 1.5 seconds.

The build currently targets only Windows (NSIS and MSI). Cross-platform builds for macOS and Linux are identified as future work (see Chapter 6).

4.11 Performance Observations

Informal timing measurements were taken during testing to assess NFR-04 (Responsiveness). All measurements were made on the Windows 11 test machine described above.

Table 4.2: Informal response-time observations for key operations.

Operation	Tauri
Application cold start (to UI ready)	≈ 1.5 s
Login (local identity + MongoDB pull)	<500 ms
Task list load (100 tasks, SQLite)	<100 ms
Task create (SQLite write + SignalR event)	<80 ms
First AI token rendered (General mode)	≈ 400 ms
Offline-to-online sync replay (20 entries)	≈ 1.8 s
Direct message delivery (sender to recipient)	<150 ms

All interactive operations satisfied the 200 ms threshold defined in NFR-04, with the exception of application cold start and AI first-token latency, which are dominated by process initialisation and network round-trip time respectively and are not considered interactive UI responses.

Chapter 5

Conclusion

This chapter summarises the work carried out, evaluates the extent to which the original objectives were achieved, reflects on the technical decisions made during the project, and acknowledges the limitations of the delivered system.

5.1 Summary of Work

TaskFlow was conceived in response to a concrete problem: existing task management tools are web-first services that become inaccessible or functionally degraded when network connectivity is lost, yet mobile or remote workers routinely face exactly those conditions. The objective was to build a self-contained desktop application that delivers the full feature set of a modern task management platform while remaining fully operational without an internet connection and automatically synchronising data when connectivity is restored.

The six objectives established in Section 1.3 were all met:

1. **Cross-platform desktop application.** A Tauri v2 shell embeds the ASP.NET Core backend and the React 19 frontend in a single redistributable package. The application packages into a Windows installer of approximately 35 MB with a cold-start time of approximately 1.5 seconds. Installers were produced for Windows (NSIS and MSI). No prior installation of Node.js, .NET, Rust, or any other runtime is required on the target machine (Section 4.10).
2. **Offline-first task management.** Every write operation produces both an immediate SQLite record and a `SyncOutboxEntry` in the outbox queue. The `OfflineSyncService` replays the queue when connectivity is restored and `BulkSyncStartupService` handles the case in which the user was offline during the preceding session. Twenty-entry offline-then-sync scenarios were verified without data loss (Section 4.6).

3. **Multiple task visualisation modes.** Five views — Default, Kanban, Table, Gantt, and Calendar — each present the task dataset through a different organisational lens. The Kanban drag-and-drop, the Gantt year timeline, and the Calendar weekly time-grid were all implemented and tested on at least two operating systems (Section 4.4).
4. **Team collaboration and real-time communication.** Project and team invitations flow through MongoDB as a relay layer. Direct messages and group chats are persisted in SQLite and delivered in real time via a SignalR hub. File attachments up to 20 MB are supported. Notification delivery was verified for all seventeen event types (Sections 4.5 and 4.9).
5. **Integrated AI assistant.** The chatbot integrates the Mistral API in three modes: general productivity guidance, code generation and explanation, and document OCR and summarisation. Responses are streamed token-by-token over Server-Sent Events with a median first-token latency of approximately 400 ms in the general mode (Section 4.7).
6. **Maintainable, extensible codebase.** The backend follows a layered architecture (Controllers, Services, Repositories, Data) enforced by constructor injection via the ASP.NET Core DI container. The `ISyncableEntity` abstraction decouples the sync engine from individual entity types, allowing new syncable entities to be added by implementing the interface and registering a corresponding repository. AutoMapper profiles and FluentValidation validators are co-located with their respective DTOs and entities.

5.2 Technical Contributions

The project produced five technical contributions that go beyond assembling existing libraries.

Outbox-pattern offline sync engine for a desktop .NET application. While the transactional outbox pattern is well documented in distributed-systems literature, applying it within a single-user SQLite-backed desktop application required adapting the pattern to a process-local context. The `ConnectivityService` acts as the circuit breaker; the `OfflineSyncService` provides the consumer loop; and the `ISyncableEntity` interface defines the contract that entities must satisfy to participate in sync. The result is a lightweight, dependency-free sync engine that requires no message broker and adds less than one millisecond of overhead per write operation on the test machine.

Dual-database architecture with a shared entity abstraction. Combining an embedded relational database (SQLite) as the primary store with a cloud-hosted document database (MongoDB) as a relay layer is an uncommon design for a desktop application. The `ISyncableEntity` interface exposes a `ToMongoDocument()` method that produces a BSON-compatible dictionary, keeping the transformation logic on the entity itself rather than

scattered across service classes. This design made it straightforward to add or modify syncable entities throughout development without changing the sync engine.

SSE-streamed multi-modal AI chatbot in a desktop context. Integrating a streaming HTTP response (SSE) inside a Tauri-hosted application required ensuring that the native WebView did not buffer or terminate the long-lived HTTP connection before all tokens were delivered. The solution — issuing the SSE request directly from the React renderer process to the local ASP.NET Core server — bypassed the shell entirely and delivered correct streaming behaviour.

Self-contained desktop packaging with embedded runtime. Bundling an ASP.NET Core server, a React frontend, and a Tauri shell into a single installer that starts all three processes transparently on Windows required careful orchestration of the Tauri main process. The stdout-signal protocol (`TASKFLOW_BACKEND_READY:{address}`) decouples the frontend load from the backend start time, allowing the UI to display a splash state until the server is confirmed ready regardless of machine speed.

Tauri sidecar architecture with splashscreen lifecycle management. The Tauri v2 [5] desktop shell packages the ASP.NET Core backend as a sidecar binary [9], launched and managed by Rust code in `backend.rs`. A two-window lifecycle (undecorated splashscreen during backend startup, then main window with the full application) provides visual feedback without requiring the user to understand the multi-process architecture. The splashscreen closes atomically when the backend signals readiness via stdout, ensuring the UI is never displayed before the API is available. Nine IPC commands bridge the Rust shell and the web renderer, covering backend URL discovery, startup status, password-reset code retrieval, window management, and application metadata. Neither the React + TypeScript frontend nor the ASP.NET Core backend required any modification to accommodate the desktop shell, confirming the framework-agnostic design of the three-tier architecture. The resulting Tauri-based installer was approximately 35 MB in size with a cold-start time of approximately 1.5 seconds on the test machine (see Table 4.2).

5.3 Lessons Learned

Several broader lessons emerged from the design and implementation process.

Offline-first requires deliberate UI design, not just backend changes. It is not sufficient to enqueue writes; users need visible confirmation that their actions have been recorded and will eventually reach the server. The `ConnectivityBar` with its pending-count badge, colour-coded connectivity states, and animated sync progress bar was as important to the user experience as the outbox queue itself. Implementing this required defining a clear contract

between the sync service and the React component so that both could react to the same state transitions.

Choosing the right conflict resolution strategy early saves significant rework. The decision to use last-write-wins based on `UpdatedAt` was made early and shaped the entity interface, the outbox entry schema, and the MongoDB upsert logic. Had this decision been deferred, adding a timestamp field and revision counter to every entity late in the project would have required a database migration on potentially live data.

SignalR group-per-user is a reliable notification pattern. Joining each authenticated connection to a group named after the user's ID (`Groups.AddToGroupAsync(Context.ConnectionId, userId)`) meant that notifications could be dispatched to any service class by calling `_hubContext.Clients.Group(userId)`, without needing to track connection IDs. This pattern handled reconnection transparently: a reconnecting client re-joined its group and received any notifications dispatched after reconnection without any server-side state changes.

Sidecar stdout parsing requires line-buffered input handling. The stdout-signal approach (`TASKFLOW_BACKEND_READY`) works well in production builds but can be disrupted during development when the backend writes diagnostic output before the ready signal. Filtering the stdout stream with a prefix rather than reading the first line proved more robust, but a dedicated named-pipe or socket protocol would be a more principled solution.

Clean architectural boundaries enable shell-layer substitution. Because the architecture separates the desktop shell from the web-based UI and the process-based backend (Section 3.2), the Rust source files in `src-tauri/` and the Tauri build pipeline are completely decoupled from the React frontend and ASP.NET Core backend. This validates the principle that investing in clean boundaries at the start of a project pays dividends when technology choices need to be revisited.

5.4 Limitations

The following limitations were identified during development and testing.

Last-write-wins conflict resolution. The synchronisation engine uses a simple last-write-wins strategy based on the `UpdatedAt` timestamp. In scenarios where two users edit the same shared task concurrently while offline, the later sync will silently overwrite the earlier one. No merge dialogue or conflict notification is presented to the user. For the single-user scenario (one user across multiple devices) this is acceptable; for multi-user collaborative editing it may result in lost updates.

Sub-task backend not implemented. The Table View renders a placeholder sub-task row beneath each task, and the task entity schema includes a comment referring to sub-tasks. How-

ever, no `SubTask` entity, migration, service, or API endpoint was created. The feature exists at the UI layer only and is not persisted. This is the largest gap between the stated scope and the delivered system.

Mistral API key is hardcoded. The API key used to authenticate requests to the Mistral API is stored as a compile-time constant in `MistralChatService.cs`. This violates the principle of separating secrets from source code and means that rotating the key requires a recompile and redistribution of the application. The recommended remedy — reading the key from the `TASKFLOW_MISTRAL_KEY` environment variable at startup and falling back to a configuration value — was identified but not implemented within the project timeline.

No automated end-to-end test suite. Testing was performed manually as described in Section 3.7. While the architecture is amenable to unit testing (stateless services, injected repositories), no unit tests, integration tests, or end-to-end tests were written. The absence of an automated regression suite means that confidence in correctness relies entirely on manual verification.

macOS Gatekeeper requires a manual override on first launch. Because the macOS distribution was not submitted to Apple for notarisation, users on macOS are required to right-click and select *Open* on the first launch. This is a distribution-process limitation rather than a functional defect but represents a friction point for non-technical users.

Chapter 6

Future Work

The delivered system meets all stated objectives and is fully functional as a cross-platform desktop application. However, several directions for improvement and extension were identified during development. These are grouped into short-term improvements that address known gaps in the current implementation and longer-term enhancements that would expand the scope of the system.

6.1 Short-Term Improvements

Sub-task backend implementation. The Table View renders a placeholder sub-task row for each task, but no corresponding backend exists. Implementing sub-tasks would require adding a `SubTask` entity with a foreign key to `TaskItem`, a database migration, a `SubTasksController`, and a `SubTaskRepository`. The `ISyncableEntity` interface would then be applied to `SubTask` so that sub-tasks participate in the offline synchronisation pipeline automatically. The UI work — expanding the Table View row and exposing CRUD controls — is largely already scaffolded. This is the highest-priority item because it is the largest gap between the stated scope and the delivered system.

Mistral API key security. As noted in Section 5.4, the Mistral API key is currently a compile-time constant. The immediate remedy is to read the key from an environment variable (`TASKFLOW_MISTRAL_KEY`) at startup, falling back to the `appsettings.json` configuration value, and ultimately defaulting to the embedded constant only as a last resort. In a production distribution the key should be provisioned by the installer or entered by the user on first launch and stored in the operating system’s credential store (Windows Credential Manager, macOS Keychain, or `libsecret` on Linux).

Automated test suite. The absence of automated tests is the most significant quality risk in the codebase. A practical test suite would consist of three layers:

1. **Unit tests** (xUnit) for pure service methods and the sync engine logic, using mocked repository interfaces.
2. **API integration tests** using `WebApplicationFactory<Program>` to spin up an in-memory ASP.NET Core server backed by an in-memory SQLite database, covering all controller endpoints.
3. **End-to-end tests** using Playwright to exercise the packaged desktop application, verifying that key user journeys (login, create task, go offline, create task, go online, confirm sync) complete correctly.

Adding this suite would allow regressions to be detected automatically and would provide the confidence required to accelerate future feature development.

Per-field conflict resolution. The current last-write-wins strategy resolves sync conflicts at the document level: the document with the later `UpdatedAt` timestamp overwrites the earlier one entirely. A more granular approach would track a per-field version vector, merging non-conflicting field changes and presenting a conflict dialogue only when the same field was edited concurrently. This would be a meaningful improvement for multi-device users who edit different properties of the same task on different machines while offline.

Cross-platform Tauri builds. The current Tauri build pipeline targets only Windows (NSIS and MSI). Extending the build to macOS and Linux would restore cross-platform coverage. The Rust source code in `src-tauri/` is already platform-independent; the main effort lies in configuring the build pipeline for each target triple (`aarch64-apple-darwin`, `x86_64-unknown-linux-gnu`) and adapting the `BUILD_WINDOWS.bat` script into a cross-platform CI pipeline. Once macOS and Linux builds are available, Apple notarisation would eliminate the Gatekeeper manual-override step described in Section 4.10.

6.2 Long-Term Enhancements

Mobile companion application. A React Native application sharing the MongoDB relay layer would allow users to view and update tasks on iOS and Android devices. Because the MongoDB layer already stores a current snapshot of every syncable entity, the mobile client would not require an additional backend service: it could connect directly to MongoDB (via Atlas Data API or a lightweight proxy) and write changes that the desktop client would pick up on its next sync cycle.

AI-powered task creation from natural language. The existing Mistral integration could be extended to accept a free-text description such as *“Set up a team meeting next Thursday at 10 AM to review the sprint”* and return a structured `CreateTaskDto` or `CreateCalendarEventDto` payload that is immediately submitted to the appropriate API endpoint. This would reduce the number of form fields a user must complete for routine task creation and use the AI assistant to speed up workflows beyond question answering.

CRDT-based synchronisation. Replacing last-write-wins with a Conflict-free Replicated Data Type (CRDT) would allow concurrent edits from multiple devices to be merged deterministically without data loss. A suitable approach for the task entity would be to model each editable field as a Last-Write-Wins Register (LWW-Register) with a logical clock, producing merge results that are commutative, associative, and idempotent. This would require replacing the current `UpdatedAt`-based merge in `UserDataSyncService` with a per-field CRDT merge function, and storing the logical clock alongside each field in the MongoDB document.

Analytics and reporting dashboard. The current dashboard provides aggregate counts and a seven-day task trend. A richer analytics view would include burndown charts, task completion velocity per team member, average task age by priority, and project-level progress bars. These metrics could be computed with MongoDB aggregation pipelines over the relay collection, avoiding additional load on the local SQLite database.

Third-party integration plugins. A plugin API that allows third-party integrations to be registered as background services would extend the application’s reach. High-value initial integrations would include bidirectional synchronisation with GitHub Issues (mapping issues to tasks), Google Calendar (mapping `CalendarEvents` to Google Calendar entries), and Slack (posting task assignment and due-date notifications to a channel). Each plugin would implement a common `IIntegrationPlugin` interface, keeping the core application decoupled from external service details.

Cloud-hosted multi-tenant mode. The current architecture is single-user and desktop-only. A future SaaS variant would replace the embedded Tauri desktop shell with a web-hosted frontend, the per-user SQLite database with a per-organisation PostgreSQL instance managed by Entity Framework Core, and the MongoDB relay with a shared cluster. The ASP.NET Core service layer would require minimal changes because its business logic is already decoupled from the storage layer via repository interfaces; the main effort would be in the authentication layer (multi-tenant JWT issuance, organisation scoping on all queries) and the deployment pipeline (containerisation, orchestration, secrets management).

Appendix

Appendix A

API Reference

Table A.1 lists all REST API controllers, their base routes, and the primary endpoints they expose. All routes are prefixed with `/api`. Every endpoint except the authentication endpoints and the connectivity check requires a valid `X-User-Email` header, resolved by the local-identity middleware described in Section 3.4.

Table A.1: REST API controllers and primary endpoints.

Controller	Base route	Key endpoints
AuthController	/api/auth	POST /register, POST /login, POST /forgot-password, POST /reset-password
TasksController	/api/tasks	GET, POST /, GET, PUT, DELETE /{id}, PATCH /{id}/status, PATCH /{id}/star, GET /search
ProjectsController	/api/projects	GET, POST /, GET, PUT, DELETE /{id}, POST /{id}/invite, GET /{id}/members
TeamsController	/api/teams	GET, POST /, GET, PUT, DELETE /{id}, POST /{id}/invite, POST /invitations/{id}/accept
MessagesController	/api/messages	GET /conversations, GET /{userId}, POST /, DELETE /conversation/{userId}
GroupChatsController	/api/groupchats	GET, POST /, GET, PUT, DELETE /{id}, POST /{id}/members, POST /{id}/messages
NotificationsController	/api/notifications	GET /, GET /unread-count, PATCH /{id}/read, PATCH /read-all, DELETE /{id}
CalendarEventsController	/api/calendarevents	GET, POST /, GET, PUT, DELETE /{id}
ChatbotController	/api/chatbot	GET /conversations, POST /conversations, GET /conversations/{id}/messages, POST /conversations/{id}/messages, POST /scan
DashboardController	/api/dashboard	GET /stats, GET /due-this-week, GET /recent-activity
TaskCommentsController	/api/tasks/{taskId}/comments	GET, POST /, DELETE /{id}
SettingsController	/api/settings	GET, PUT /profile, PUT /password
ConnectivityController	/api/connectivity	GET /ping, POST /bulk-sync
DevController	/api/dev	GET /db-info (development environment only)

Appendix B

Database Schema

SQLite Entities

Table B.1 lists all eighteen SQLite entities managed by Entity Framework Core, together with their primary key type, key fields, and any sync participation.

Table B.1: SQLite entity summary.

Entity	PK	Key fields
AppUser	int Id	FirstName, LastName, Email, PasswordHash, AvatarUrl, Company, Country, Phone, Timezone, ResetToken, ResetTokenExpiry
TaskItem	int Id	Title, Description, Status, Priority, DueDate, IsStarred, ProjectId, AssigneeId, SyncId, UpdatedAt, IsSynced
Project	int Id	Name, Description, Color, OwnerId, IsStarred, SyncId, UpdatedAt, IsSynced
ProjectMember	int Id	ProjectId, UserId, Role, JoinedAt
Team	int Id	Name, Description, OwnerId, CreatedAt
TeamMember	int Id	TeamId, UserId, Role, JoinedAt
Message	int Id	SenderId, ReceiverId, Body, IsRead, SentAt, AttachmentUrl, IsDeletedBySender, IsDeletedByReceiver, IsEdited
GroupChat	int Id	Name, CreatedByUserId, CreatedAt; child: GroupChatMember (GroupChatId, UserId, JoinedAt); child: GroupChatMessage (GroupChatId, SenderId, Body)
Notification	int Id	UserId, Title, Message, NotificationType, IsRead, ActionUrl, RelatedTaskId, SyncId, UpdatedAt
CalendarEvent	int Id	OwnerId, Title, Description, StartAt, EndAt, Color, MeetingLink
Chatbot Conversation	int Id	UserId, Title, Mode, CreatedAt, UpdatedAt
ChatbotMessage	int Id	ConversationId, Role, Content, CreatedAt
Reminder	int Id	TaskId, UserId, FireAt, NotifyEmail, NotifyInApp, HasFired, FiredAt, SyncId
SyncOutboxEntry	int Id	OperationId, OperationName, PayloadJson, Status (Pending/Processing/Synced/Failed), Attempts, LastAttemptAt, ErrorMessage
TaskComment	int Id	TaskId, AuthorId, Body, CreatedAt
LocalInvitation	int Id	InviteeId, InviterId, TeamId or ProjectId, Status, CreatedAt
LocalTeamMember	int Id	TeamId, UserId, Role, JoinedAt (local mirror of MongoDB record)
ISyncableEntity	—	Interface only: SyncId, UpdatedAt, IsSynced, ToMongoDocument()

MongoDB Collections

Table B.2 lists the MongoDB collections used as the relay and mirror layer (fifteen in total).

Table B.2: MongoDB collection descriptions.

Collection	Description and key fields
<code>user_presence</code>	Optional online-status documents. Fields: <code>userId</code> , <code>lastSeenAt</code> , <code>isOnline</code> .
<code>team_invitations</code>	Cross-device invitation relay. Fields: <code>inviterEmail</code> , <code>inviteeEmail</code> , <code>teamId</code> , <code>projectId</code> , <code>teamName</code> , <code>status</code> (Pending/Accepted/Declined), <code>createdAt</code> .
<code>team_members</code>	Mirror of accepted team memberships. Fields: <code>teamId</code> , <code>userId</code> , <code>userEmail</code> , <code>role</code> , <code>joinedAt</code> .
<code>cross_notifications</code>	Cross-device notification relay for multi-machine alerts. Fields: <code>recipientEmail</code> , <code>title</code> , <code>message</code> , <code>notificationType</code> , <code>relatedTaskId</code> , <code>createdAt</code> .
<code>user_accounts</code>	Encrypted password-hash backup for account restoration after reinstall. Fields: <code>email</code> , <code>passwordHash</code> , <code>userId</code> , <code>backedUpAt</code> .
<code>team_announcements</code>	Team-wide announcement storage. Fields: <code>teamId</code> , <code>authorEmail</code> , <code>title</code> , <code>message</code> , <code>createdAt</code> , <code>readBy</code> .
<code>app_settings</code>	Key-value configuration store. Documents keyed by <code>_id</code> .
<code>users</code>	Mirror of <code>AppUser</code> profile data (Company, Country, Phone, Timezone). Fields: <code>email</code> , <code>fullName</code> , <code>avatarUrl</code> , <code>company</code> , <code>country</code> , <code>phone</code> , <code>timezone</code> .
<code>teams</code>	Mirror of <code>Team</code> entity. Fields: <code>name</code> , <code>description</code> , <code>color</code> , <code>ownerId</code> , <code>createdAt</code> .
<code>team_member_records</code>	Mirror of per-member team metadata (distinct from <code>team_members</code>). Fields: <code>teamId</code> , <code>userId</code> , <code>role</code> , <code>joinedAt</code> .
<code>messages</code>	Mirror of <code>Message</code> entity. Fields: <code>senderId</code> , <code>receiverId</code> , <code>text</code> , <code>fileUrl</code> , <code>mimeType</code> , <code>fileSize</code> , <code>readAt</code> , soft-delete flags.

Collection	Description and key fields
<code>chatbot_conversations</code>	Mirror of AI chatbot conversation records. Fields: <code>userId</code> , <code>title</code> , <code>mode</code> (General/Coder/Scanner), <code>createdAt</code> .
<code>chatbot_messages</code>	Mirror of individual AI chat messages. Fields: <code>conversationId</code> , <code>role</code> (user/assistant), <code>content</code> , <code>createdAt</code> .
<code>tasks</code>	Mirror of <code>TaskItem</code> . Documents keyed by <code>syncId</code> (GUID). Fields: <code>syncId</code> , <code>title</code> , <code>description</code> , <code>status</code> , <code>priority</code> , <code>dueDate</code> , <code>ownerId</code> , <code>projectId</code> , <code>isStarred</code> , <code>updatedAt</code> .
<code>notifications</code>	Mirror of <code>Notification</code> . Fields: <code>syncId</code> , <code>userId</code> , <code>title</code> , <code>message</code> , <code>notificationType</code> , <code>isRead</code> , <code>actionUrl</code> , <code>relatedTaskId</code> , <code>updatedAt</code> .

Appendix C

Configuration Reference

`appsettings.json` Structure

The application is configured via `appsettings.json` (overridden by `appsettings.Development.json` in development). Table C.1 describes each configuration section.

Table C.1: appsettings.json section reference.

Section / key	Description
DatabaseProvider	"SQLite" (default) or "SqlServer". Selects the EF Core provider at startup.
ConnectionStrings. DefaultConnection	SQLite connection string, e.g. Data Source=taskflow.db. Relative paths are resolved against the executable directory.
MongoDB. ConnectionString	Full MongoDB connection string including credentials. Leave empty to run in SQLite-only mode without sync.
Jwt.Key	HMAC-SHA256 signing key (minimum 32 characters) (<i>vestigial</i> — <i>the local-identity middleware does not issue or validate JWT tokens</i>).
Jwt.Issuer / .Audience	Token issuer and audience strings (<i>vestigial</i>).
Jwt.ExpiresInMinutes	Token lifetime in minutes (<i>vestigial</i>).
Mistral.Model	Model ID for the General chatbot mode. Default: mistral-small-latest.
Mistral.CoderModel	Model ID for the Coder mode. Default: codestral-latest.
Mistral.OcrModel	Model ID for the Scanner (OCR) mode. Default: mistral-ocr-latest.
Mistral.HistoryWindow	Number of prior messages included in the Mistral context. Default: 20.
EmailSettings.*	SMTP configuration for MailKit. Smtphost, Smtpport (587), UseSsl (false for STARTTLS), SenderEmail, SenderName, Username, Password.
Kestrel.Endpoints. Http.Url	http://127.0.0.1:0 — binds to a random available port. The chosen port is printed to stdout as TASKFLOW_ BACKEND_READY:{address}.

Environment Variables

Table C.2 lists the environment variables recognised by the application at startup.

Table C.2: Supported environment variables.

Variable	Effect
ASPNETCORE_ENVIRONMENT	Standard ASP.NET Core environment selector. Set to <code>Development</code> to enable Swagger UI and verbose logging.
ASPNETCORE_URLS	Overrides Kestrel endpoint binding when set. Normally not set; the <code>Kestrel</code> config section is used instead.

tauri.conf.json Summary

Table C.3 summarises the key `tauri.conf.json` settings for the Tauri-based installer.

Table C.3: Key `tauri.conf.json` configuration (Tauri build).

Key	Value / description
<code>productName</code>	<code>TaskFlow</code>
<code>identifier</code>	<code>com.taskflow.desktop</code>
<code>build.frontendDist</code>	<code>../wwwroot</code> — directory containing <code>index.html</code> and Webpack output bundles.
<code>app.windows[0]</code>	Main window: 1280×800 , centred, initially hidden, show on backend-ready signal.
<code>app.windows[1]</code>	Splashscreen: 400×300 , undecorated, initially hidden, shown during backend startup.
<code>app.security.csp</code>	Content Security Policy allowing localhost API and SignalR WebSocket connections.
<code>bundle.targets</code>	<code>["nsis", "msi"]</code> — Windows installer formats.
<code>bundle.externalBin</code>	<code>["binaries/taskflow-backend"]</code> — sidecar path.
<code>bundle.windows.webviewInstallMode</code>	<code>downloadBootstrapper</code> — download WebView2 runtime if absent.
<code>bundle.windows.nsis.installMode</code>	<code>currentUser</code> — no administrator privileges required.
<code>plugins.shell.open</code>	<code>true</code> — permits opening URLs in the default browser.

Bibliography

- [1] David Allen. *Getting Things Done: The Art of Stress-Free Productivity*. Penguin Books, 2001.
- [2] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7B. arXiv preprint arXiv:2310.06825, 2023. Available: <https://arxiv.org/abs/2310.06825>.
- [3] Marc Shapiro, Nuno Preguiça, Carlos Baquero, and Marek Zawirski. Conflict-Free Replicated Data Types. In *Proceedings of the 13th International Symposium on Stabilization, Safety, and Security of Distributed Systems (SSS)*, pages 386–400. Springer, 2011.
- [4] David J. Anderson. *Kanban: Successful Evolutionary Change for Your Technology Business*. Blue Hole Press, 2010.
- [5] Tauri Contributors. Tauri v2 Documentation. <https://v2.tauri.app/start/>, 2025. Accessed 2025.
- [6] Microsoft Corporation. Microsoft Edge WebView2. <https://developer.microsoft.com/en-us/microsoft-edge/webview2/>, 2025. Accessed 2025.
- [7] Tauri Contributors. Tauri Security: Capabilities. <https://v2.tauri.app/security/capabilities/>, 2025. Accessed 2025.
- [8] The Rust Programming Language. The Rust Programming Language. <https://www.rust-lang.org>, 2025. Accessed 2025.
- [9] Tauri Contributors. Embedding External Binaries in Tauri. <https://v2.tauri.app/develop/sidecar/>, 2025. Accessed 2025.
- [10] Electron Contributors. Electron Documentation. <https://www.electronjs.org/docs>, 2024. Accessed 2024.
- [11] Martin Kleppmann. *Designing Data-Intensive Applications*. O’Reilly Media, 2017.

- [12] Eric Brewer. CAP Twelve Years Later: How the “Rules” Have Changed. *Computer*, 45(2):23–29, 2012.
- [13] Martin Fowler. *Patterns of Enterprise Application Architecture*. Addison-Wesley, 2002.
- [14] Martin Kleppmann and Alastair R. Beresford. A Conflict-Free Replicated JSON Datatype. *IEEE Transactions on Parallel and Distributed Systems*, 28(10):2733–2746, 2017.
- [15] Kevin P. Gaffney, Martin Prammer, Larry Brasfield, D. Richard Hipp, Dan Kennedy, and Jignesh M. Patel. SQLite: Past, Present, and Future. *Proceedings of the VLDB Endowment*, 15(12):3535–3547, 2022.
- [16] Kristina Chodorow. *MongoDB: The Definitive Guide*. O’Reilly Media, 2nd edition, 2013.
- [17] Microsoft. Entity Framework Core Documentation. <https://docs.microsoft.com/en-us/ef/core>, 2024. Accessed 2024.
- [18] I. Fette and A. Melnikov. The WebSocket Protocol. RFC 6455, Internet Engineering Task Force, December 2011. Available: <https://www.rfc-editor.org/rfc/rfc6455>.
- [19] Microsoft. ASP.NET Core SignalR Documentation. <https://docs.microsoft.com/en-us/aspnet/core/signalr>, 2024. Accessed 2024.
- [20] Niels Provos and David Mazières. A Future-Adaptable Password Scheme. In *Proceedings of the 1999 USENIX Annual Technical Conference*, pages 32–32. USENIX Association, 1999.
- [21] OWASP Foundation. OWASP Top 10 Application Security Risks – 2021. <https://owasp.org/Top10/>, 2021. Accessed 2024.
- [22] Ravi S. Sandhu, Edward J. Coyne, Hal L. Feinstein, and Charles E. Youman. Role-Based Access Control Models. *Computer*, 29(2):38–47, 1996.
- [23] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. *Advances in Neural Information Processing Systems*, 30, 2017.
- [24] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.
- [25] James M. Wilson. Gantt Charts: A Centenary Appreciation. *European Journal of Operational Research*, 149(2):430–437, 2003.